

OpenEHR



eHealth Master SS2024
Sten Hanke

WHAT IS OPENEHR?

- Open Standard supporting the definition, persistence and querying of structured clinical data



<https://www.openehr.org/>

- An open specification for a health information model
- capable of supporting an open platform ecosystem
 - vendor neutral
 - technology neutral
- licensed to allow open and closed source business models



<https://www.openehr.org/>

1. A methodology for **defining and curating clinical content**; and
2. The specification of an **open API and query language** that allows openEHR content to be stored and queried in a compliant Clinical Data Repository (CDR).

- **Terminologies** allowing the inclusion of any terminology such as SNOMED-CT;
- **Clinical Decision Support** through its Guideline Definition Language (GDL); and
- **Workflow and Task Planning** through its Task Planning Visual Modelling Language (TP-VML).

- Technology and vendor neutral with a number of both open source and proprietary implementations
- Maintained by a not-for-profit openEHR Foundation based in London
- Large community sharing clinical content definitions
- Is aligned with international standards like ISO/TR 20514, ISO/TS 18308, HL7, FHIR, SNOMED-CT, IHE-XDS

why openEHR? - 'open platform'

- must be based on a common, open information model
 - open specification
 - open source clinical content definitions
- other parts of the ecosystem need not be open source
- critical infrastructure for new app developers
 - substantially faster development
 - much lower barriers to market
 - no vendor / technology lock-in

Entity	Organisation metrics	Country
 Queensland Health	citizens: 4000000	Australia 
 Northern Territory Health	citizens: 233000	Australia 
Ministry of Health Malta		Malta 
 HelseVest	citizens: 1000000	Norway 
 Nasjonal IKT, Norway	healthcare practitioners: 137000 hospitals: 24	Norway 
 Moscow City Department of Information Technologies	clinics: 780 healthcare practitioners: 130000	Russia 
Region Oestergoetland		Sweden 
 Ministry of Health, Slovenia	citizens: 2000000	Slovenia 

<https://inidus.com/who-has-adopted-open-platforms/>

Gartner.
WHY GARTNER ANALYSTS RESEARCH EVENTS CONSULTING ABOUT

Evaluate openEHR Standards for Managing Clinical Content Across the Care Continuum

Published: 07 April 2017 ID: G00317087

Analyst(s): [Mike Jones](#) | [Henrik Hallenberg](#)

Summary

openEHR standards offer a potential solution to help HDO CIOs retain control of care delivery data in the long term. New capabilities built using these standards can provide a foundation for federated care record services and innovation, but will require investment in clinical informatics skills.

<https://www.gartner.com/doc/3672617/evaluate-openehr-standards-managing-clinical>

openEHR

Mary Wilkins
July 1, 1976 (38y 5m) • F
1234567

Kardio • N3C-11/1 • 14-Nov-2014 • 12
2008 Chronic heart disease
Acute exacerbation

Reference weight: 100 kg

ATC APL

azide 500 mg
DOSE 1-1-2-2 mg — 4X per day — When needed — Intravenous

Clarithromycin (Clarithromycin)
DOSE 500 mg — 2X per day — Oral

Dopamine (Dopamine 200mg/5ml solution for infusion pre-filled syringes)
RATE 10 ml/h — DOSE 2.5 mg/kg/h — CONTINUOUS INFUSION — Intravenous

Gliclazide (Gliclazide)
DOSE 120 mg — Morning — Oral

1001051 Miller, Chris
DOB 27/06/1944 Walker (9876543)
Discharged: 15/09/2011

Suspend until:

Activities Pathology List Patient ADT Procedures/Devices IV Devices eICAT data

Activity	Status	Start	End	Scheduled	MDT
BSI data - Enterococcus faecium (VRE)	Active	15/09/2011			
Communication log	Proposed				
Contact trace data - VRE_092011	Proposed				
HBCIS Alert	Completed		15/09/2011		
Isolate Patient	Completed		15/09/2011		
Notify Nurse Unit Manager	Completed		15/09/2011		
Perform serology	Scheduled			29/09/2011	
SO data - Enterococcus faecium (VRE)	Proposed				

Update Mobile Sta

Informasjon om pasienten

SIGMUNDSTAD*, Ramiza*

23.09.1985*47516 • 28 år • Mann
Stafloren 84, 9016 Tromsø
9103009
anonym@post.no

Case og kritisk informasjon
Annet: Ulaget stoff - Anonymisert tekst
Smitte Til informasjon: Blodsmitte

Revisjonsdrift
Anonymisert tekst
Andre aktive primærhenvisninger: 2
[LA TROMSØHOSP]

Pasientens diagnose
F20.9 Ulagert schizofreni
F15.1 Skadelig bruk av andre stimulanter, inkl. 0 (18)
F20.1 Rebellert schizofreni
184 (0)

Poliklinikk 20-Aug-2013 kl 12:00
PSTO - Voksnepsykiatri Tromsø

20-Aug-2013
Anonymisert tekst
PSTO - Voksnepsykiatri Tromsø

02-Jul-2013
Anonymisert tekst
PSTO - Voksnepsykiatri Tromsø

Tidligere sykdommer
hepatitt C 2003
Sist endret 1. feb. 4. 2003

Medisinsk koding
Ingen koder registrert på kontakten

IDCR POC

Search...

Q Search

Demonstration

Back Patients Details

Ivor Cox Gender: Male Born: 06-Jun-1944 NHS No.: 9999999000

Address: 6948 Et St., Haiesowen, Worcestershire, VX27 5DV

Phone: (011961) 32362

GP Details: Goff, Carolyn D.

PAS No.: 352541

Results

Filter...

Test Name	Sample Taken	Date Created	Source
Urea, electrolytes and creatinine measurement	21-Feb-2015	21-Mar-2015	openehr
Urea, electrolytes and creatinine measurement	22-Feb-2015	22-Mar-2015	openehr

Result

Test Name: Urea, electrolytes and creatinine measurement

Status: Final

Sample Taken: 21-Feb-2015

Results

Result	Value	Unit	Normal Range
Urea	7.4	mmol/l	
Creatinine	127.0	mmol/l	
Sodium	135.0	mmol/l	
365760004	5.3	mmol/l	

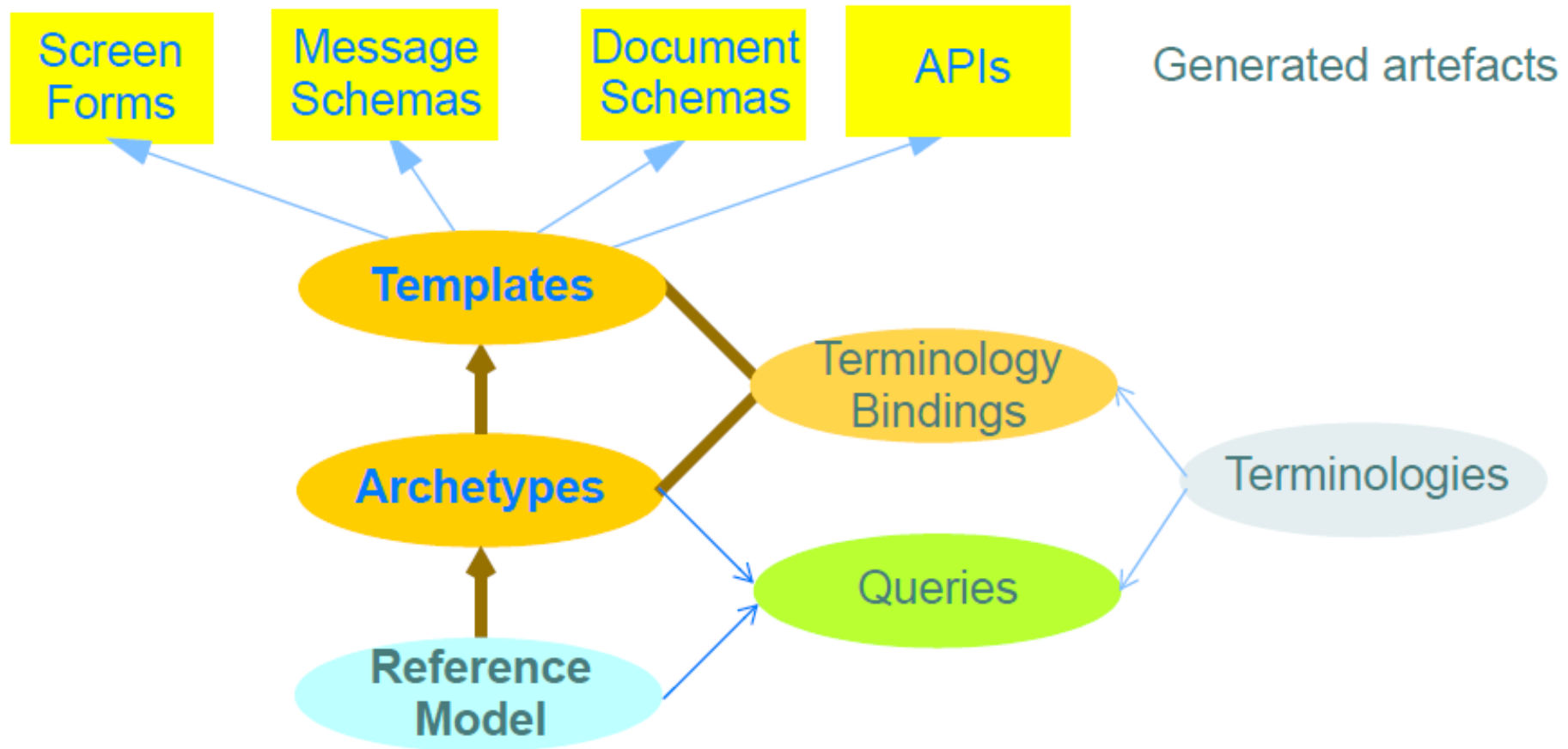
Conclusion: Rapidly deteriorating renal function

- clinical content defined using openEHR is stored on Clinical Data Repositories (CDR)
- OpenEHR clinical models can also be used in other modalities such as HL7 FHIR or RDF (Resource Description Framework)

	Database	Vendor-neutral	AQL	GDL support	open source	Separate product
Think!EHR	Oracle / SQL Server / PostgreSQL	Yes	Yes	Yes		Yes
OceanEHR	SQL server	Yes	Yes	Yes		Yes
DipsEHR	SQL Server	Yes	Yes	Yes		Yes
EtherCIS	PostgreSQL	Yes	Yes	In dev	Yes	Yes
Base24	PostgreSQL	Yes	In dev	In dev		Yes
Cabolabs	Any SQL	Yes	In dev	Nn dev	Yes	Yes
Privantis	PostgreSQL	Yes	In dev	In dev		In dev
Ehr.care	GraphDBs/Postgres	Yes	Yes	In dev		Yes
Clever CDR		Yes	Yes			Yes

- Open **governance model** encourages standardisation while supporting local variations
- **Designed for clinicians** as the domain experts to build clinical models
- Robust **modelling methodology** to cover health and social care
- **Global community** who collaborate on the creation and curation of clinical content
- **Library of open source clinical models**
- **Semantically coherent** providing computable models for developers
- **Human language agnostic**

- **Vendor and technology neutrality** – multiple proprietary and open source CDR implementations
- **Zero-engineering deployment of new content** - new openEHR models can be deployed to any CDR with no engineering (i.e. no involvement of the CDR provider, no schema changes and no data migration.)
- **Sophisticated semantic querying** - openEHRs Archetype Query Language (AQL) can be run over multiple openEHR Clinical Data Repositories irrespective of their technology and internal physical data structures.
- **Automatic versioning and audit** - openEHR CDRs maintain details of all changes and amendments to the data stored in them along with metadata to track provenance and attribution. This provides a robust medico-legally admissible audit trail with zero effort from application developers.



The architecture uses:

- **Reference model** to provide semantic coherence
- **archetypes** to define detailed elements of clinical content; and
- **templates** that combine and constrain elements from multiple archetypes to support a given use case.

Templates are used by developers to generate various artefacts that can be used to build applications. openEHR allows terminologies like SNOMED-CT to be bound to both archetypes and templates. openEHR provides a query language AQL that allows detailed querying of content stored in an openEHR CDR.

CLINICAL INFORMATION MODEL

“I want to record Pulse rate”

■ GUI

The screenshot shows the Think!Med KOOKIT condition assessment interface. At the top, the patient's name 'Wayne, Bruce' is displayed next to a cartoon icon. Below the name, patient identifiers are listed: IMCN 1402007WA0BR4, AGE 5Y 0M ♂, and N2A-01 / POST1. The main section is titled 'KOOKIT condition assessment' and contains various input fields for clinical data. These include Temperature (36 °C), Site of measurement (Buccal), Weight (10 kg = 10,000 g), Height/Length (100 cm), Rate (70 /min), Blood pressure Syst./Diast. (100 / 80 mm[Hg]), Rate (28 /min), spO2 (96 %), Pupil diameter - left and right (both set to Pin-point pupil (<= 1mm)), Consciousness - Speech (Oriented), Movement (Withdrawal response), Eyes opening (Spontaneous eye opening), Body position (45° (head) tilt), and Phase (At rest). Each input field has a corresponding '...' button to its right. At the bottom, there is a status bar showing 'MEASUREMENT bovest - ENTRY 2.3.2012 15:03' and three buttons: 'Device data', 'Confirm', and 'Cancel'.

Think!Med

Wayne, Bruce

IMCN 1402007WA0BR4 — AGE 5Y 0M ♂ — N2A-01 / POST1

KOOKIT condition assessment

Temperature 36 °C

Site of measurement Buccal

Weight 10 kg = 10,000 g 0P

Height/Length 100 cm 2P

Rate 70 /min

Blood pressure Syst. / Diast. 100 / 80 mm[Hg]

Rate 28 /min

spO2 96 %

Pupil diameter - left mm Pin-point pupil (<= 1mm)

Right mm Pin-point pupil (<= 1mm)

Consciousness - Speech Oriented

Movement Withdrawal response

Eyes opening Spontaneous eye opening

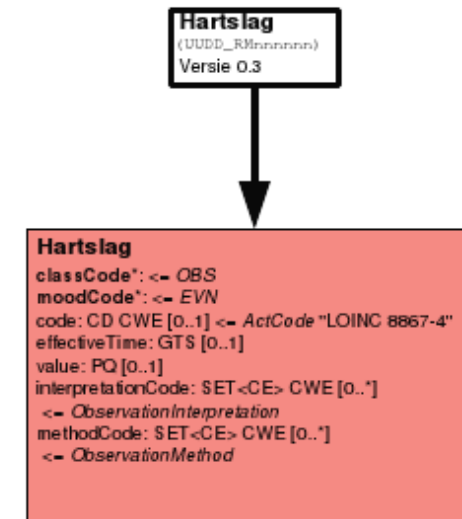
Body position 45° (head) tilt

Phase At rest

MEASUREMENT bovest - ENTRY 2.3.2012 15:03

Device data Confirm Cancel

- ▶ “What exactly do you mean by ‘Pulse rate’”?
- ▶ “Rate, rhythm, date recorded”?
- Information model (typically UML)
 - Structure (OOP classes)
 - Business rules
 - “Rate ≥ 0 ”
 - Terminology
 - SNOMED, ICD, local codes

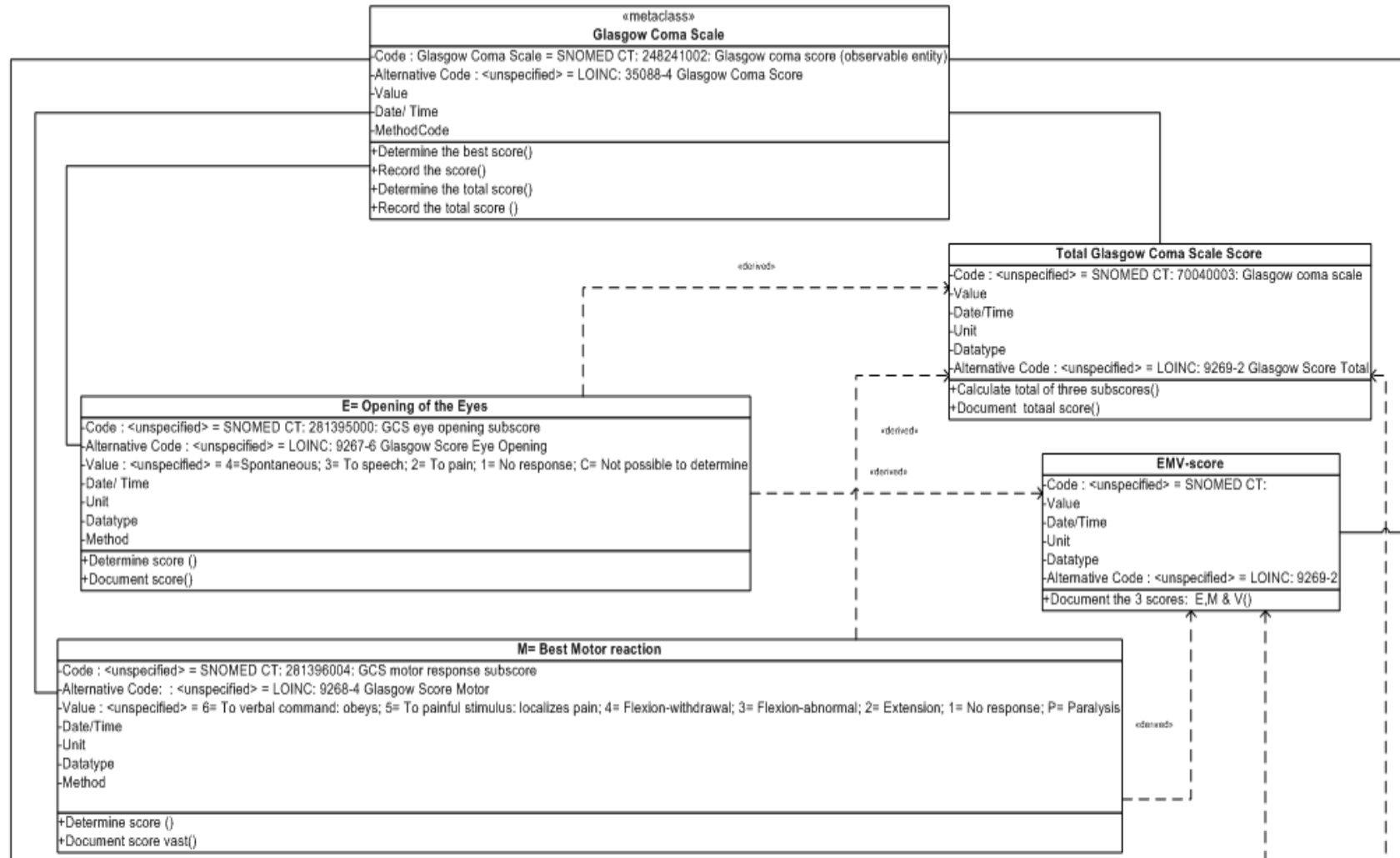


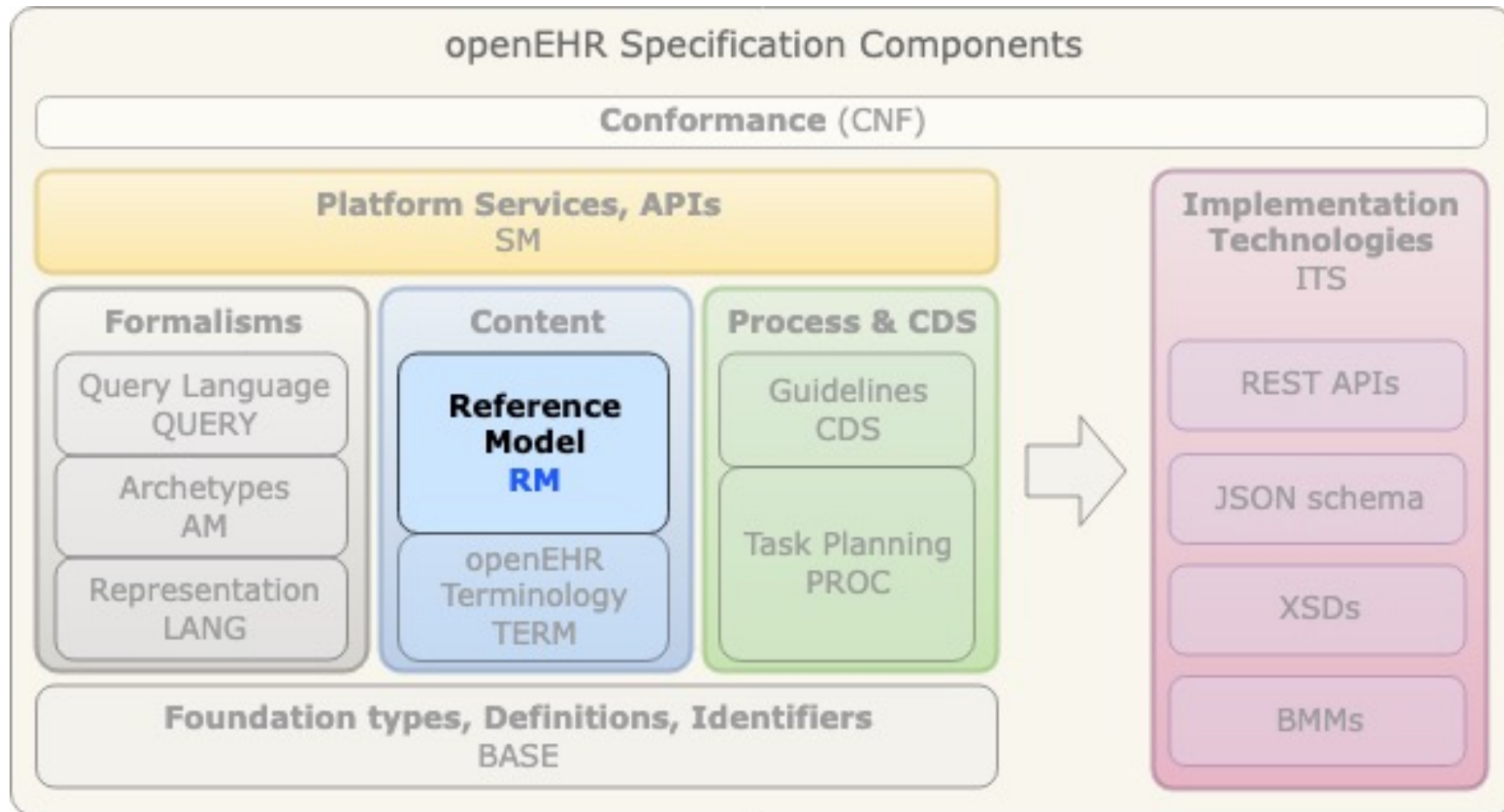
- Formally (computably) expressed in:
 - Software : Application-specific internal “Information models”, UML, database schema
 - Reference terminologies : READ, SNOMED CT, ICD-10,
 - Messaging models : HL7, CDA, IHE profiles
 - Decision support , guidelines, rules

- Informally expressed in:
 - Local forms, documents, professional guidelines
 - Data dictionaries for govt reporting

- Continually evolving: restructured, new, deprecated

Traditional modelling: UML





openEHR is underpinned by a stable **reference model**, which ensure semantic coherence providing:

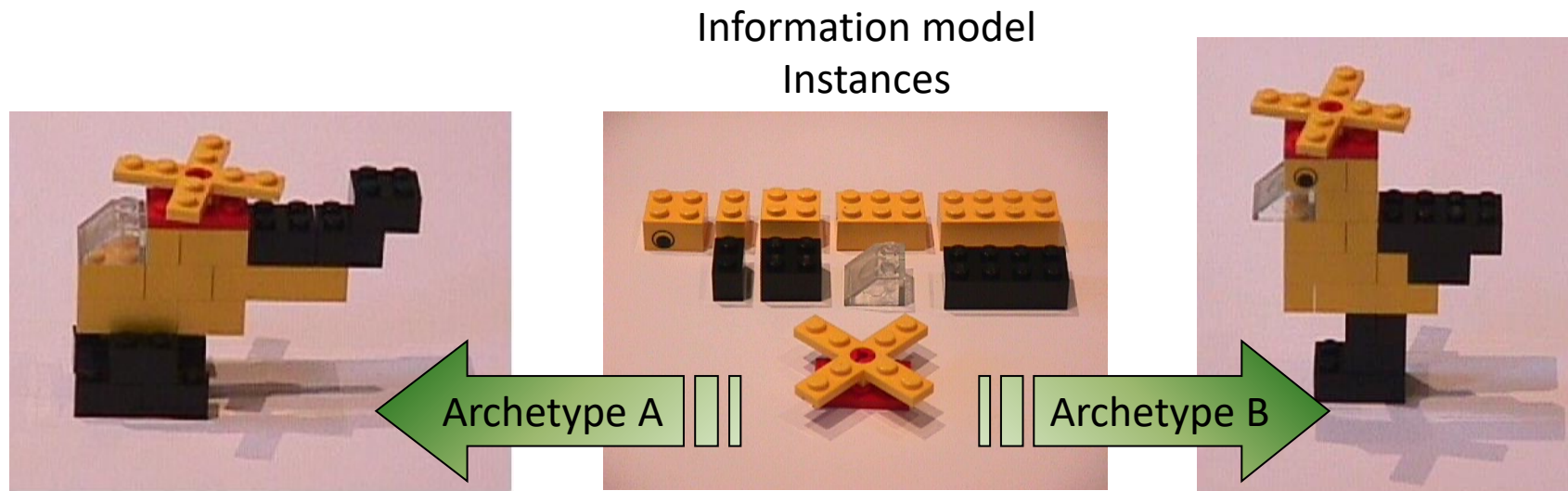
- Set of classes defining the generic structure of a patient health record:
 - Observation
 - Evaluation
 - Instruction
 - Action
 - Admin entry
- Medico-legal context and+ audit details
- Versioning
- Datatype definitions

Specifications

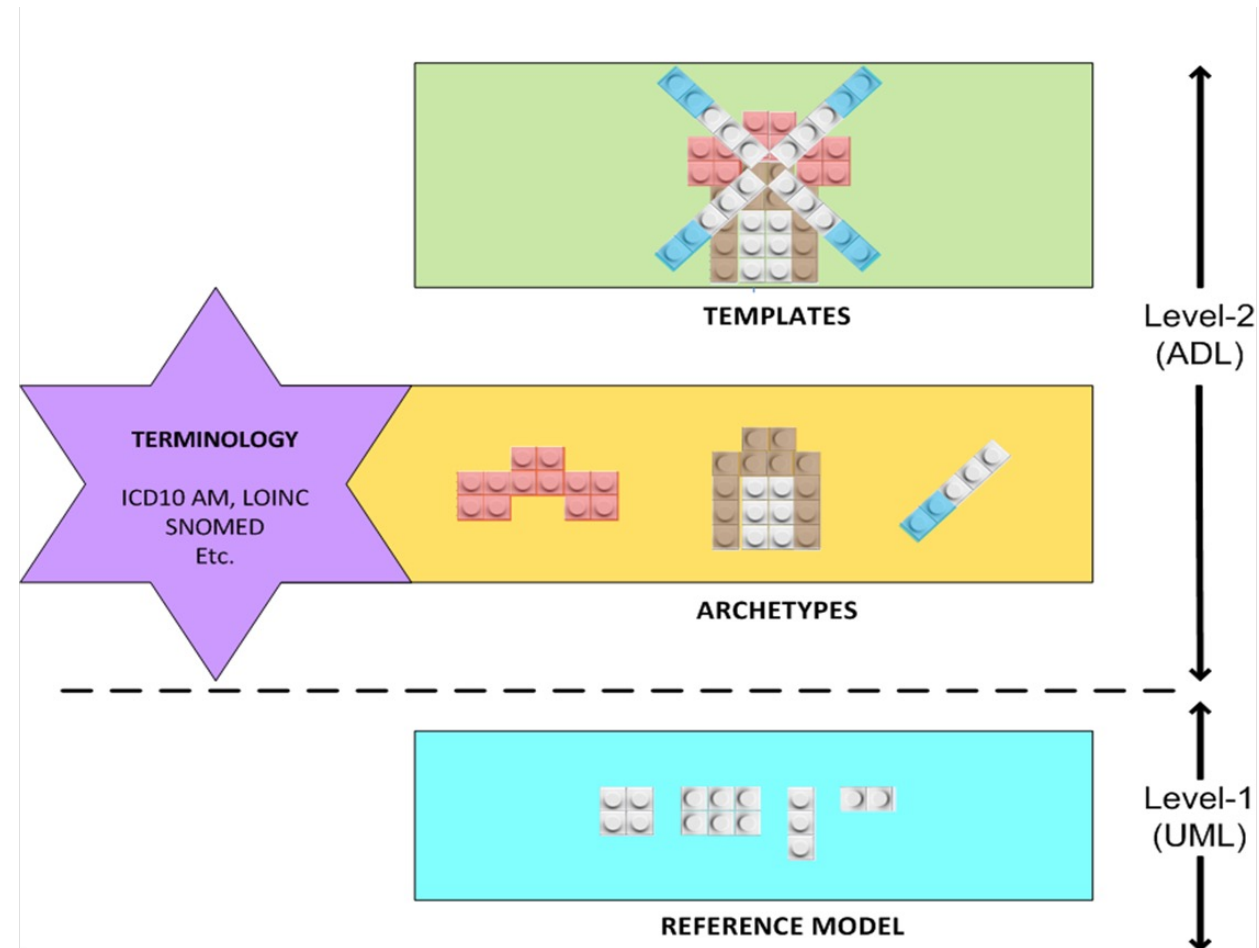
Specification	Description
STABLE EHR	The information model of the openEHR EHR.
STABLE Demographic	The openEHR demographics information model.
STABLE Common	Information model containing common concepts, including the archetype-enabling `LOCATABLE` class, party references, audits and attestations, change control, and authored resources.
STABLE Data Structures	Information model of data structures, including a powerful model of time-series data.
STABLE Data Types	Information model of data types, including quantities, date/times, plain and coded text, time specification, multimedia and URIs.
STABLE Support	Support model defining identifiers, assumed types, and terminology interface specification used in the rest of the IMs.
STABLE Integration	Information model for representing legacy data is a free-form Entry type for implementing integration solutions.
STABLE EHR Extract	The information model of the EHR Extract, which is a serialisation of content from an EHR.

- two level modelling approach based on a stable reference model. The methodology uses:
 - **archetypes** to define detailed elements of clinical content; and
 - **templates** that combine and constrain elements from multiple archetypes to support a given use case.

- The components of the Reference Model are like LEGO brick specifications
- Archetypes = instructions/designs constraining the use of LEGO pieces to create meaningful structures

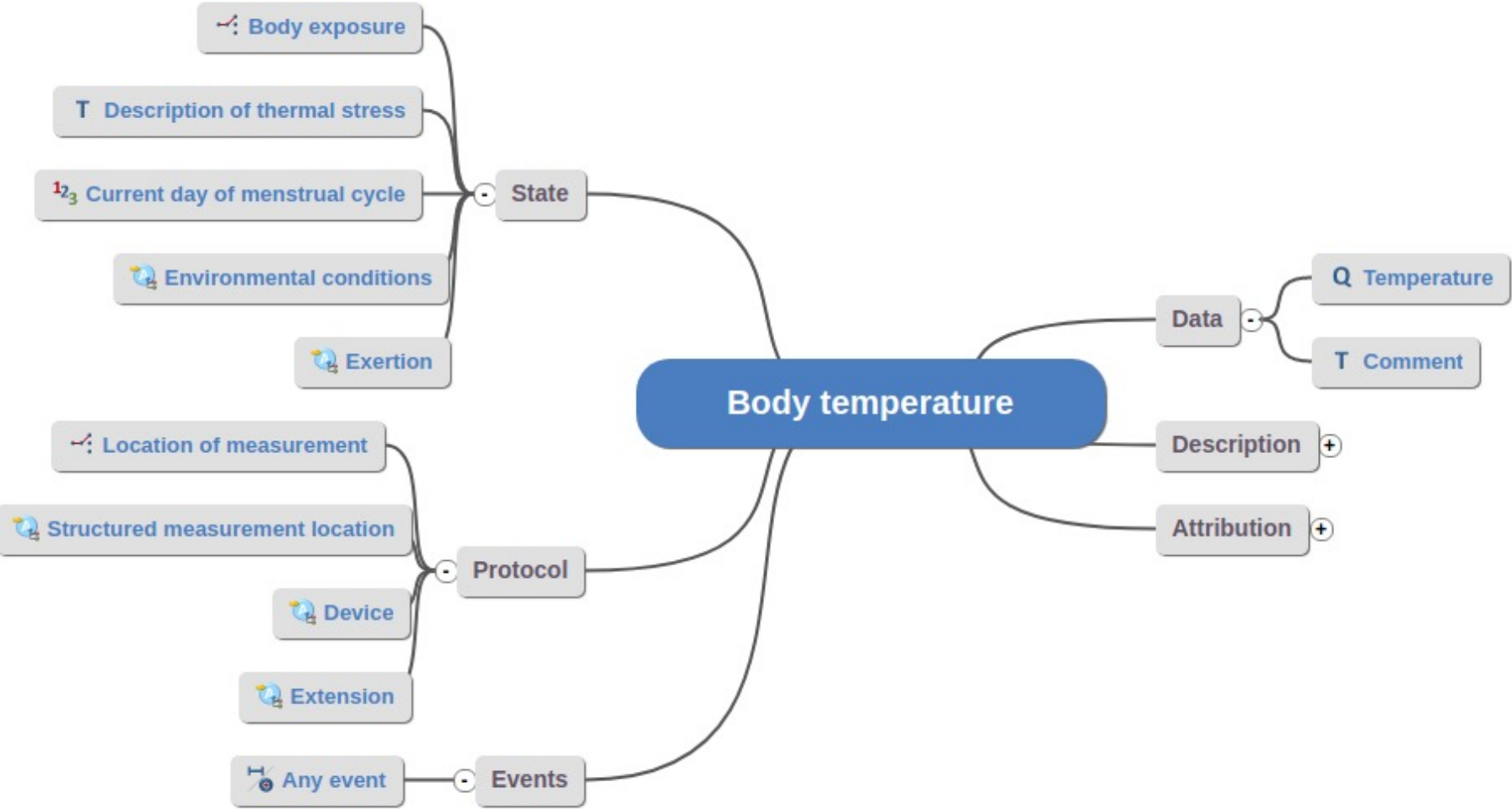


openEHR: Multi-level modelling



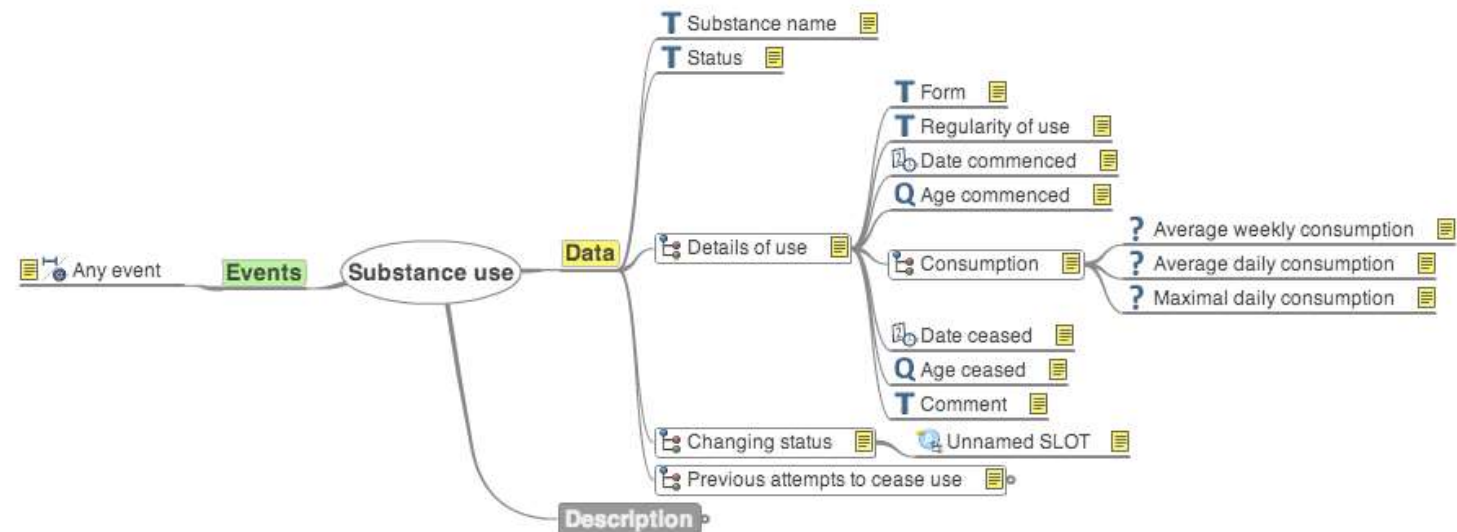
- Dictionary definition ... “a model or prototype”
- Computable models of discrete clinical concepts
 - Familiar components of a health record
 - Blood pressure, Body weight, Symptom
 - Medication order, Family history
 - Prostate cancer histopathology result
- ‘Maximal dataset’ philosophy
 - Capture as many clinical perspectives as possible
 - Universal use case

- archetype represents a piece of clinical content.
 - weight, a temperature, a blood pressure, a medication, a test or procedure
 - around 10,000 bits of clinical content that would need to be defined to cover the whole of health and social care
- openEHR archetypes are maximal data sets containing all of the data points required by all use cases
 - something simple like body temperature consists of just a value of has 12 data points plus many additional pieces of metadata.



- archetype also defines which data points are mandatory and which optional, their data type and validation criteria
- openEHR expects archetypes to be built by clinicians and other domain experts, **not** technologists
- Clinical modelling effort in the community
- network of Clinical Knowledge Manager implementations

- open source computable models of discrete clinical concepts
- Familiar components of a health record
 - Blood pressure, Body weight
 - Medication order, Family history
 - Urea, Creatinine results
- 'Maximal dataset'
 - Capture as many clinical perspectives as possible



Ocean Archetype Editor [Glasgow Coma Scale]

File Edit Language Terminology Tools Help

openEHR-EHR-OBSERVATION.glasgow_coma.v1

Header Definition Terminology Display Interface Description

☐ Protocol ☐ Participation ☐ Person State with EventSeries

Data ☐ Person State

Tree Events

☐ Ordered at0009

- Best eye response
- Best verbal response
- Best motor response
- Score

Constraint Details

Occurrences

Min: 0 Max: 1 ☐ Unbounded

Description: Best eye response to stimulus

Runtime name constraint:

Ordinal

	C	Text	Descr
1		No eye opening	*
2		Eye opening in response to pain	*
3		Eye opening to speech	*
4		Eyes opening spontaneously	*
*			

www.openehr.org/ckm

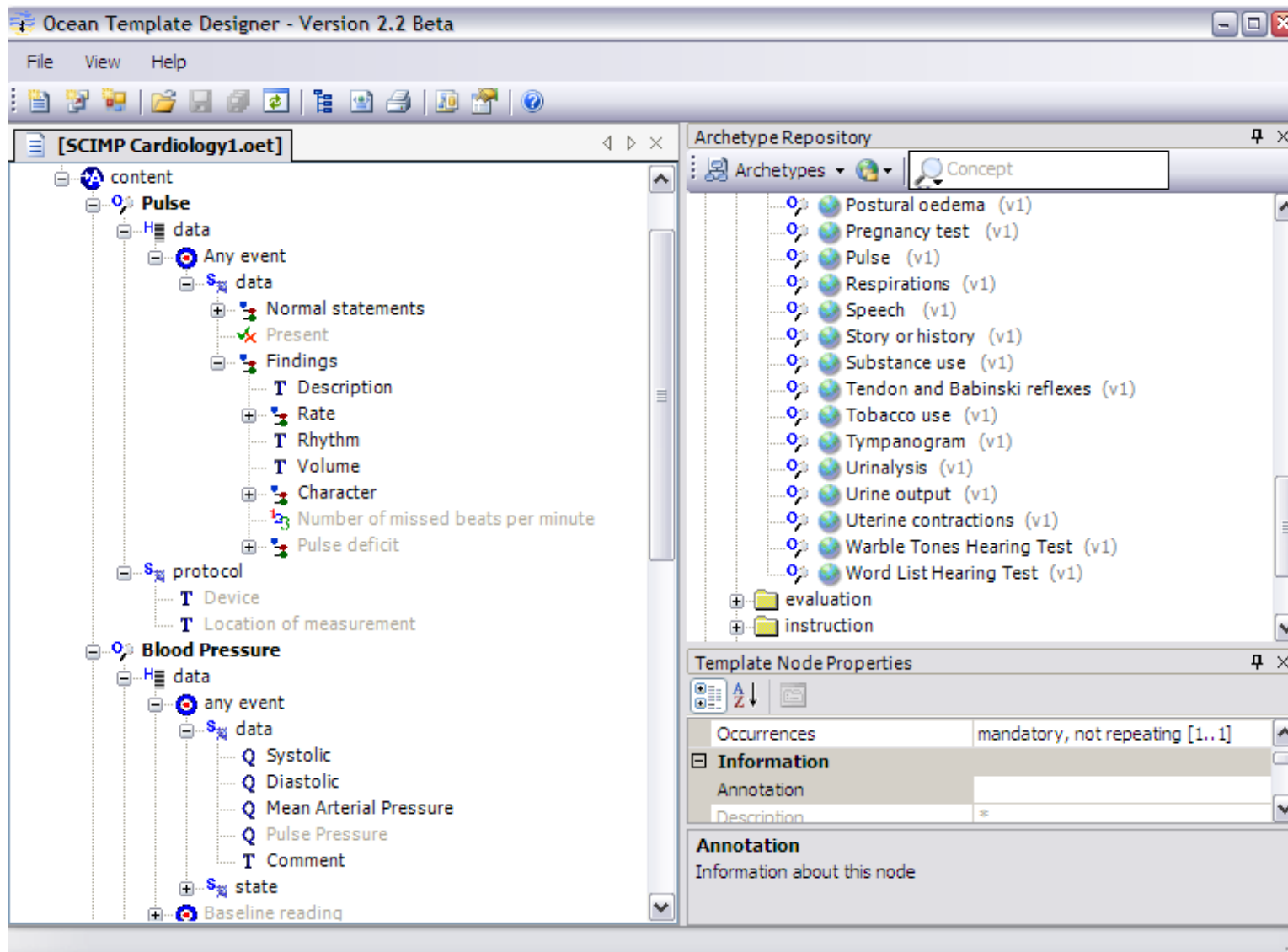
Archetype: Glasgow Coma Scale (openEHR-EHR-OBSERVATION.glasgow_coma.v1)			
Header		Data	
	Best eye response Ordinal 0..1 (<i>optional</i>)	Best eye response to stimulus	1: No eye opening 2: Eye opening in response to pain 3: Eye opening to speech 4: Eyes opening spontaneously 
	Best verbal response Ordinal 0..1 (<i>optional</i>)	Best verbal response to stimulus	1: None 2: Incomprehensible sounds 3: Inappropriate words 4: Confused 5: Oriented 
	Best motor response Ordinal 0..1 (<i>optional</i>)	Best motor response to stimulus	1: No motor response 2: Abnormal extension to pain 3: Abnormal flexion to pain 4: Flexion withdrawal from pain 5: Localizes to pain 6: Obeys commands 
	Score Count 0..1 (<i>optional</i>)	The final score	

- Archetypes get the glory
 - but templates deliver the datasets
 - Key clinical endpoint and starting point for generation of technical artefacts
 - Class libraries, GUI skeletons, Message schema

- Most demand for clinical information content will originate as requests for datasets
 - Data entry forms
 - Diabetes shared care message
 - Discharge summary message

- Further validations and terminology bindings can be added to a template in addition to those inherited from the underlying archetypes
- can be used to create document or message schemas, provide the data needed to create screens and other user interface components or define the payload of an API call

- Any appropriate terminology (e.g. SNOMED-CT, ICD, LOINC, etc.) can be integrated with openEHR models via terminology bindings
- can be bound to both openEHR archetypes and templates
- terminology bindings are generally made at the template level to allow for local variations in the use of terminology to be more easily handled



Ocean Template Designer - Version 2.2 Beta

File View Help

[SCIMP Cardiology1.oet]

content

- Pulse
 - data
 - Any event
 - data
 - Normal statements
 - Present
 - Findings
 - Description
 - Rate
 - Rhythm
 - Volume
 - Character
 - Number of missed beats per minute
 - Pulse deficit
 - protocol
 - Device
 - Location of measurement
- Blood Pressure
 - data
 - any event
 - data
 - Systolic
 - Diastolic
 - Mean Arterial Pressure
 - Pulse Pressure
 - Comment
 - state
 - Baseline reading

Archetype Repository

Archetypes Concept

- Postural oedema (v1)
- Pregnancy test (v1)
- Pulse (v1)
- Respirations (v1)
- Speech (v1)
- Story or history (v1)
- Substance use (v1)
- Tendon and Babinski reflexes (v1)
- Tobacco use (v1)
- Tympanogram (v1)
- Urinalysis (v1)
- Urine output (v1)
- Uterine contractions (v1)
- Warble Tones Hearing Test (v1)
- Word List Hearing Test (v1)
- evaluation
- instruction

Template Node Properties

Occurrences mandatory, not repeating [1..1]

Information

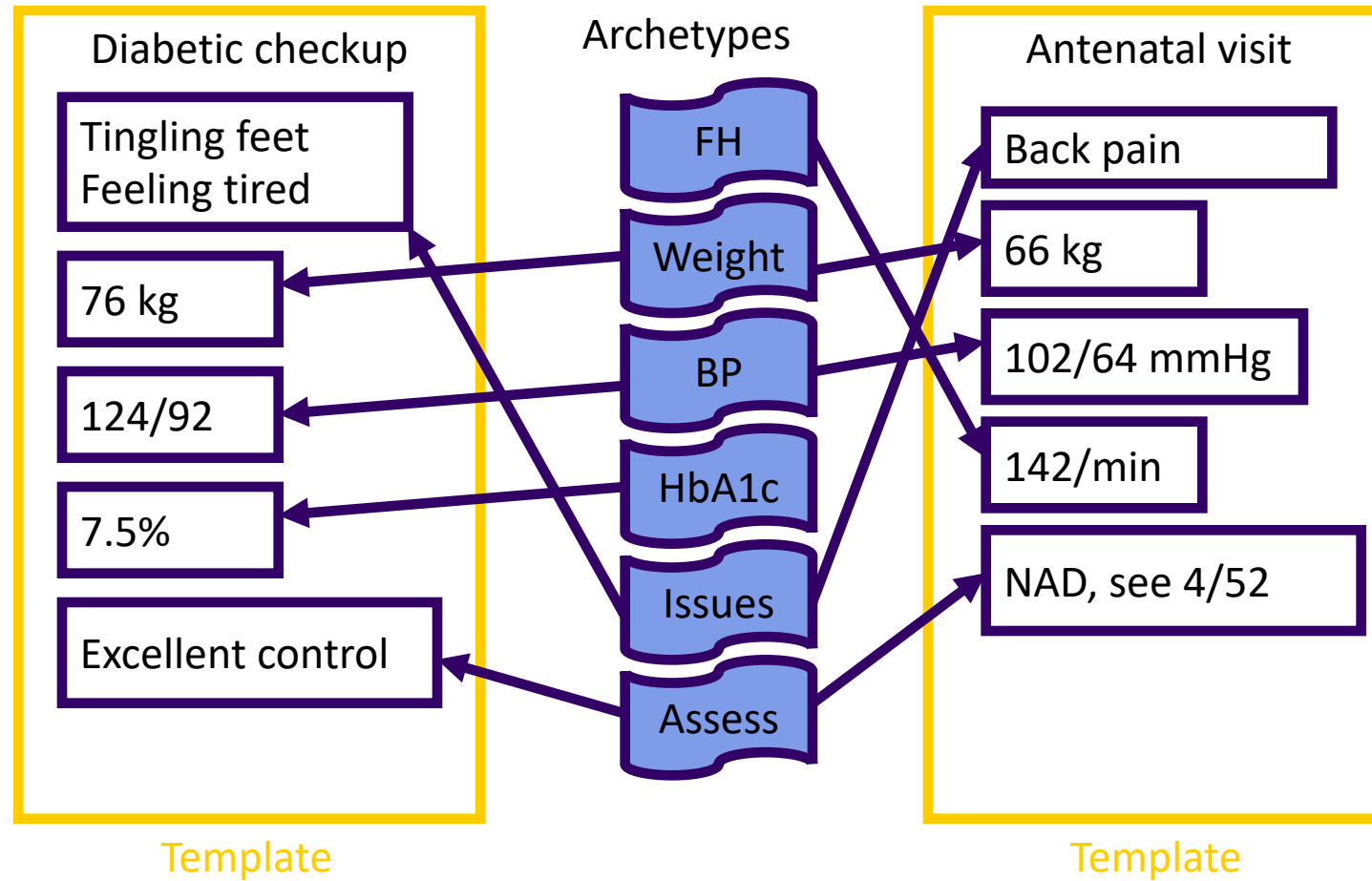
Annotation

Description *

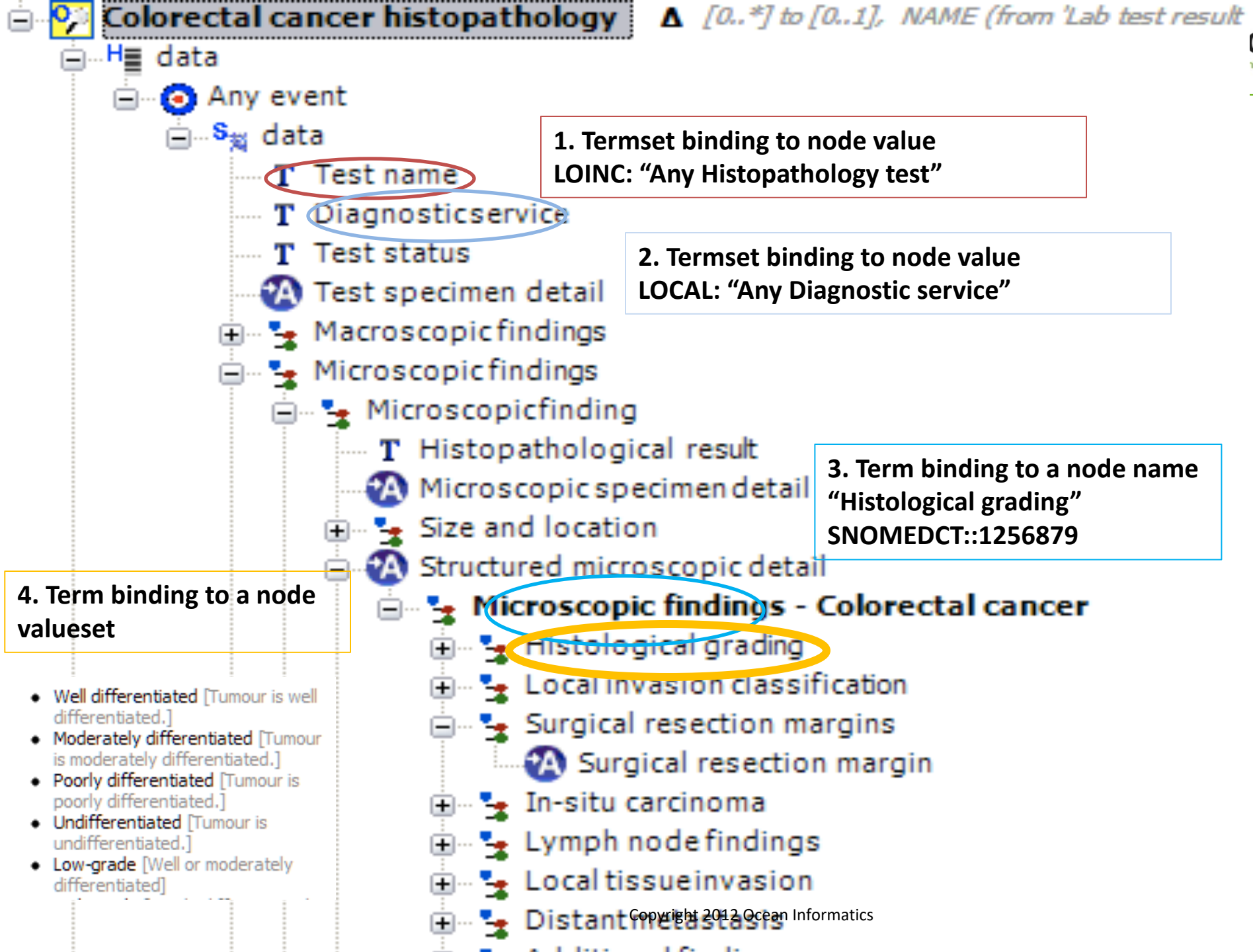
Annotation

Information about this node

Archetypes and Templates



- Each archetype carries its own unique local terminology, with multi-lingual capacity
 - “Diastolic” [at0005]
- ‘Bindings’ to multiple external terminologies
 - Define a mapping to an external term, carrying both the mapped term and local archetype term in the data



1. Termset binding to node value
LOINC: "Any Histopathology test"

2. Termset binding to node value
LOCAL: "Any Diagnostic service"

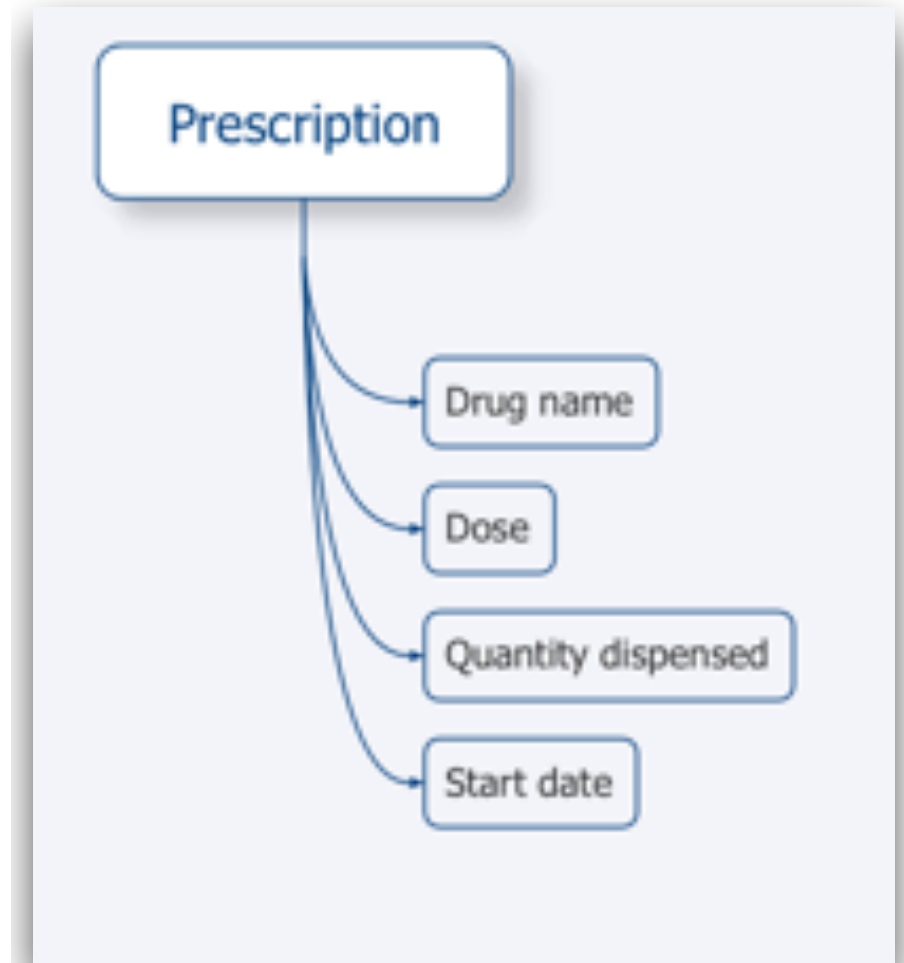
3. Term binding to a node name
"Histological grading"
SNOMEDCT::1256879

4. Term binding to a node valueset

OPENEHR CKM

- The clinical modelling tools hide most of the Reference Model and detailed understanding is not required for clinicians or 3rd party developers
- Some aspects of the RM are worth knowing as it carries a few key pieces of information that do not need to be modelled in archetypes i.e we get those 'for free'

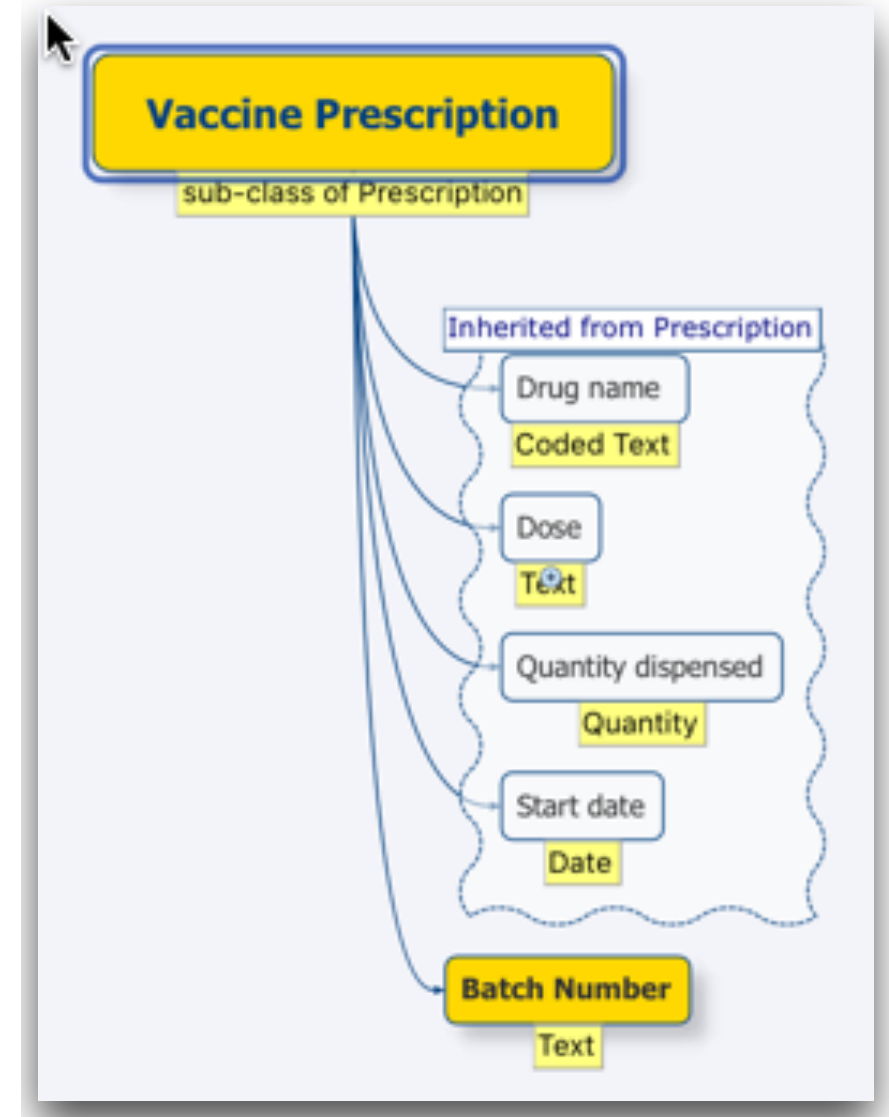
- 'Classes' are definitions of data structures
 - the 'assembly instructions' or 'recipe'
 - Classes have attributes (properties)



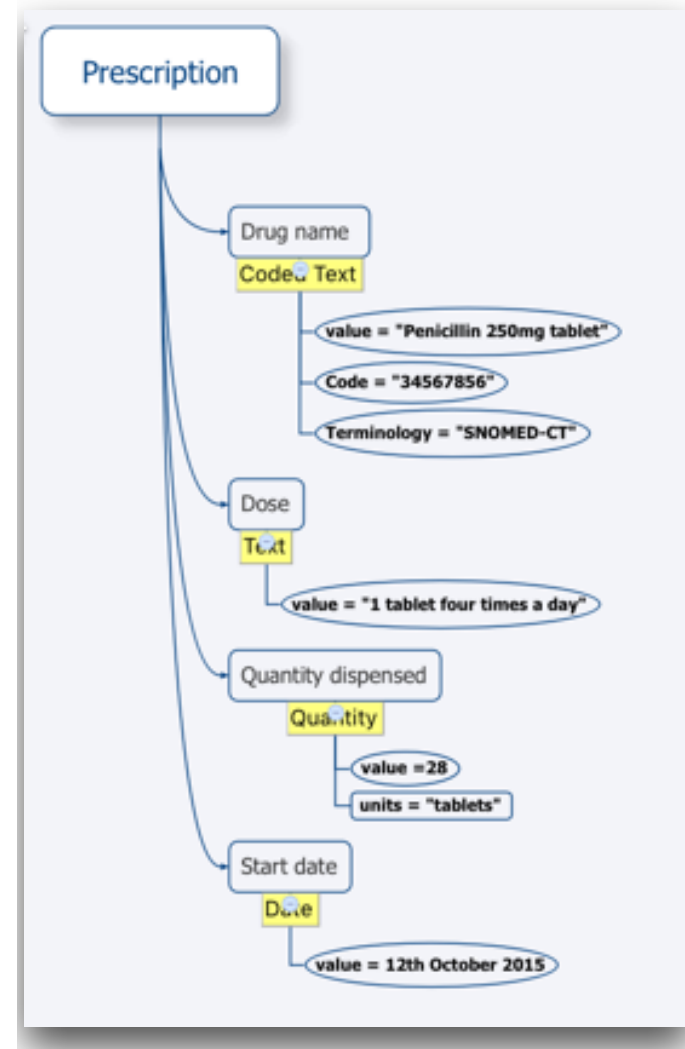
- Datatypes describe the basic **type** of information being carried
 - a piece of text
 - a quantity
 - a date or time duration an image
 - Etc., etc



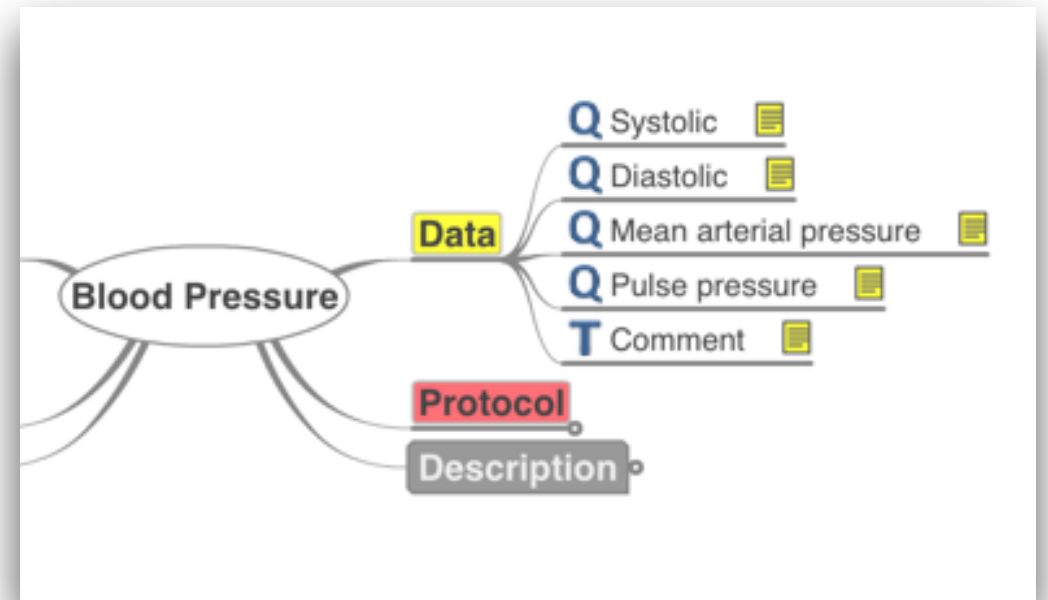
- Classes can be based on other 'parent' classes
- called inheritance or 'sub-classing'
- the sub-classes 'inherit' all the properties or attributes of the parent
 - Dog is a sub-class of Animal
 - 'Labrador' is a sub-class of 'Dog'
 - If 'Dog' has an attribute of 'tail', The Labrador class will also have 'Tail'

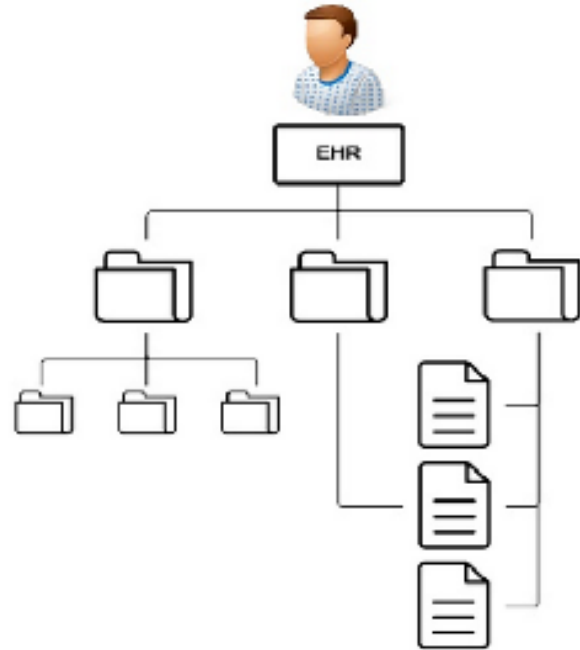


- Objects carry the data specified by the classes
 - Classes are ‘the recipe’
 - Objects are ‘the cake’

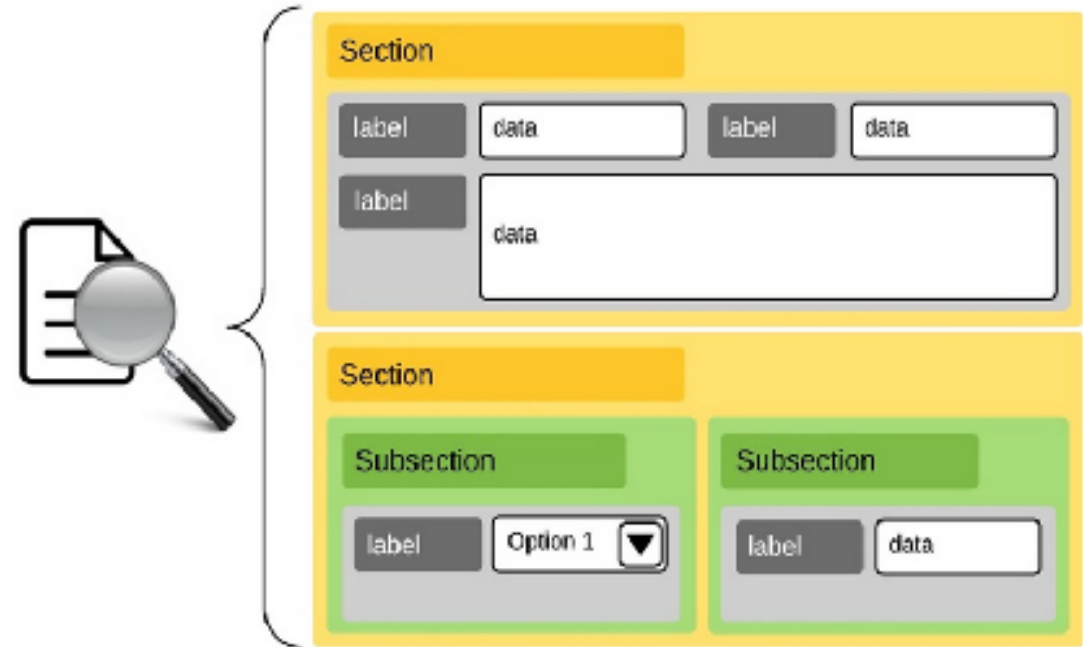


- Archetypes are built on top of the RM classes and 'inherit' their attributes
 - e.g. An Observation archetype such as Blood pressure inherits the attributes of the RM OBSERVATION class
- Archetypes use the RM datatypes
- Most of these properties are technical but some are important to clinical modellers



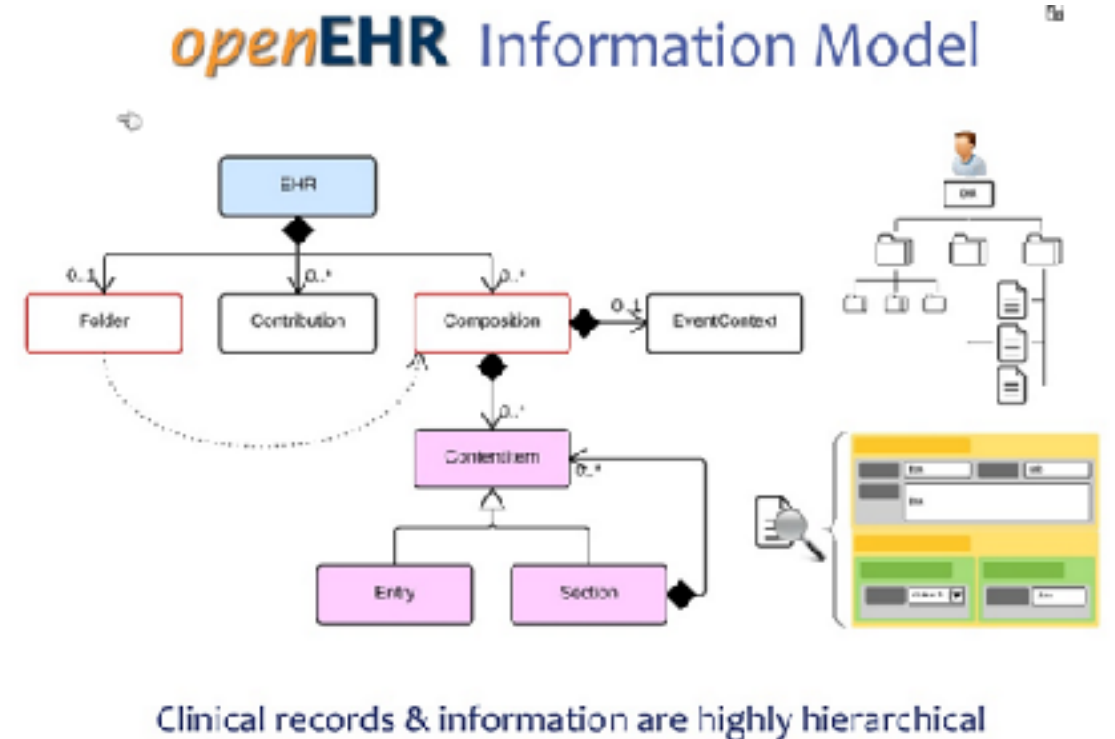


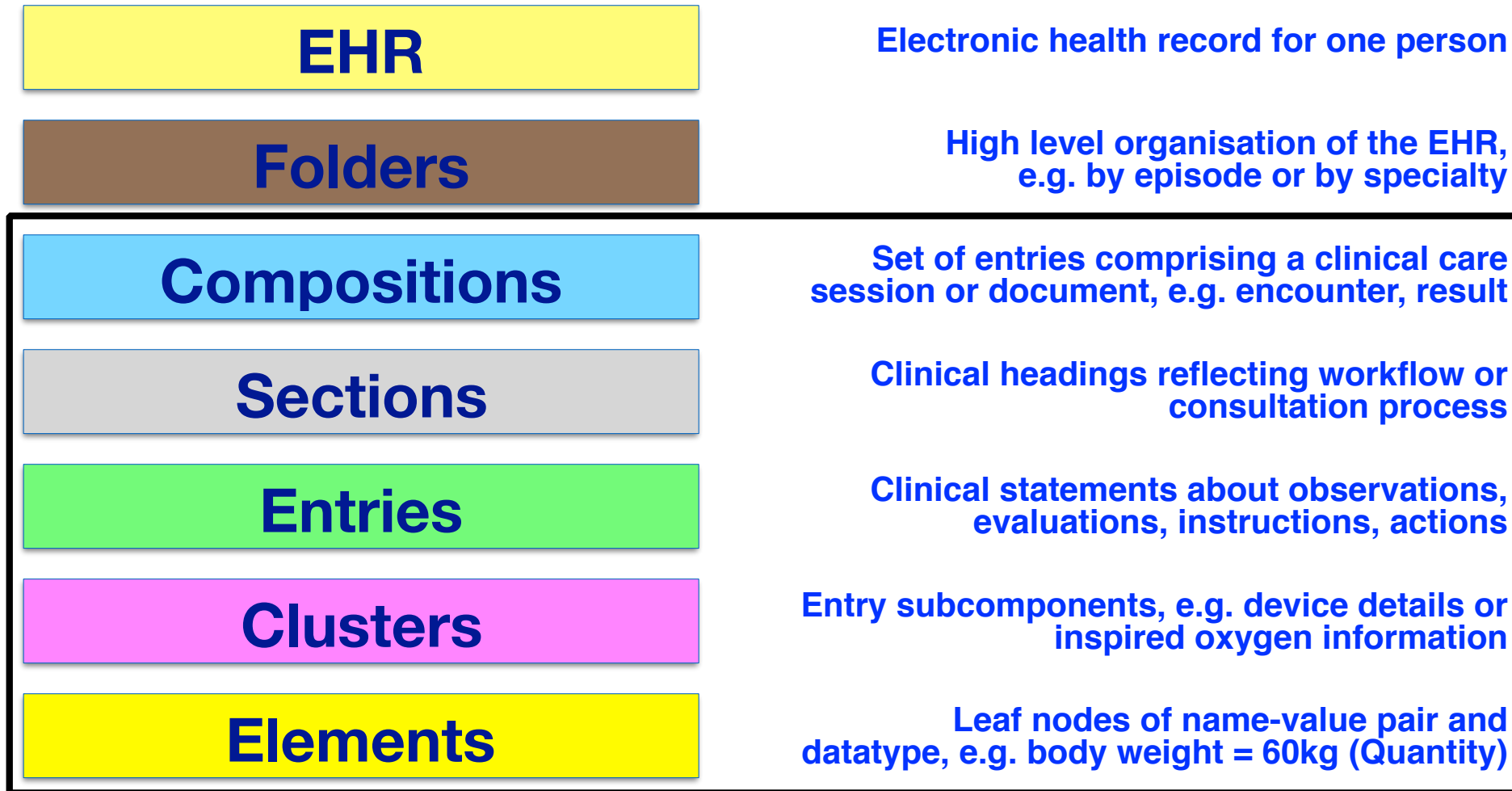
Clinical records & information are highly hierarchical
paper based or electronic

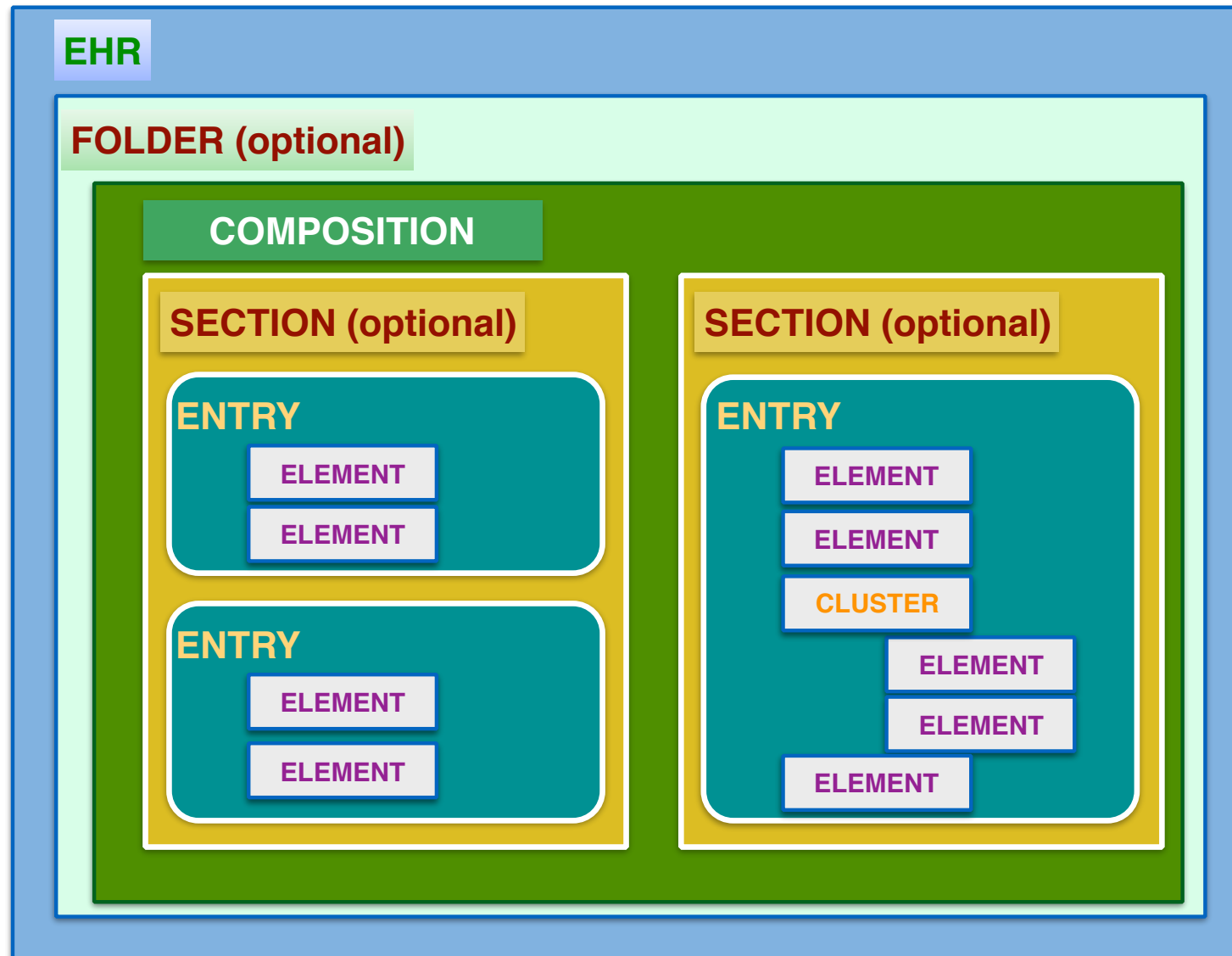


Clinical records & information are highly hierarchical

- The openEHR health record is a tree-like structure
- The root for a single patient is the **EHR**
- All of the patient data is contained in **COMPOSITIONS**
- At the end of each branch of the tree is a leaf node - the actual datapoint, the **ELEMENT** which carries the value







EHR

ehr_id = 5c8a8636-bc98-4441-abd5-e9cf396e8833

COMPOSITION = 'Nursing Observations'

SECTION Vital signs

ENTRY Blood P.

Systolic	134 mmHg
Diastolic	86 mmHg

ENTRY Pulse

Rate	134 mmHg
Rhythm	Irregular

SECTION = 'Other obs'

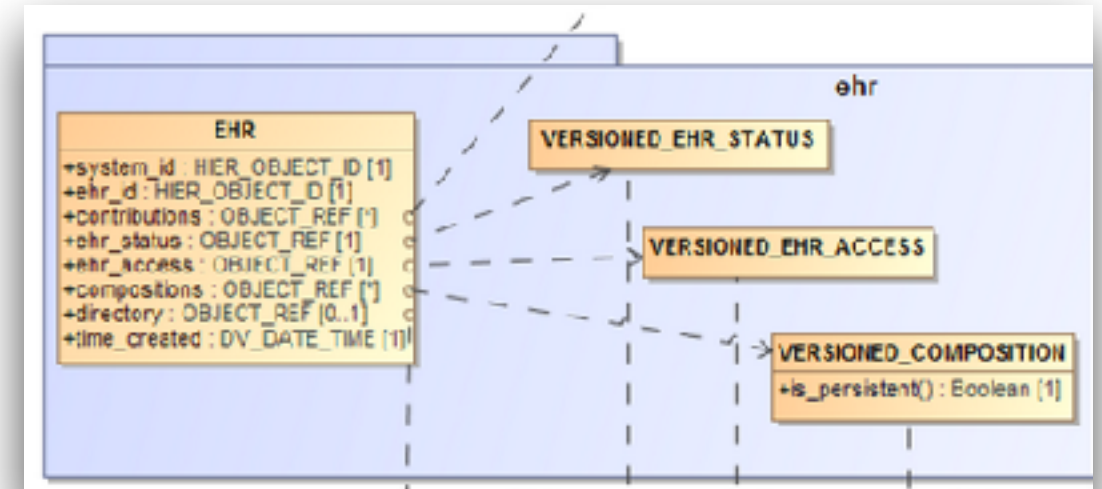
General appearance

Skin colour	Jaundiced
Behaviour	Normal

Hydration

Skin turgor	Reduced
Capillary Return	30s

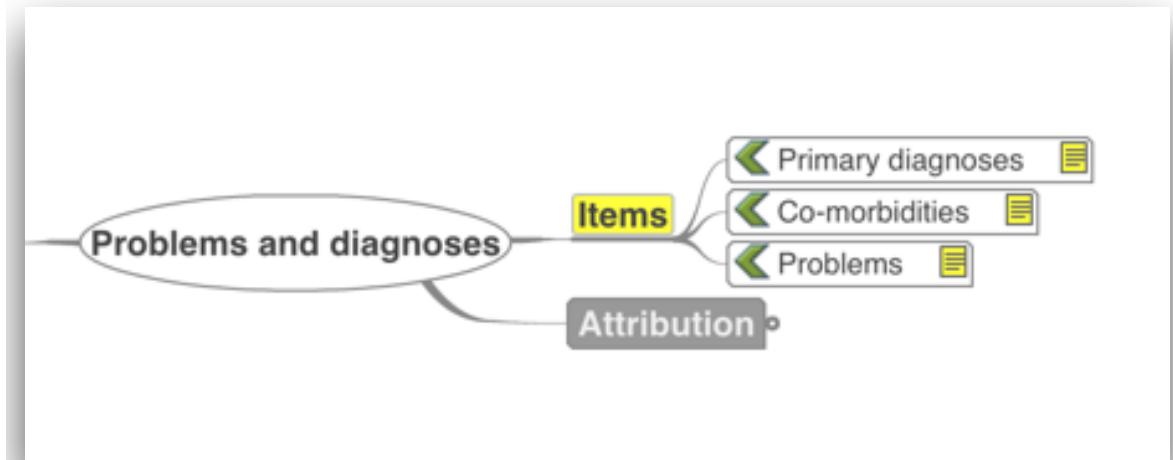
- Top-level container for all of the data for a **single patient**
- Each EHR has a unique, anonymous ID the `ehr_id`
- This needs to be associated with a real-world identifier e.g NHS Number to allow the patient to be identified



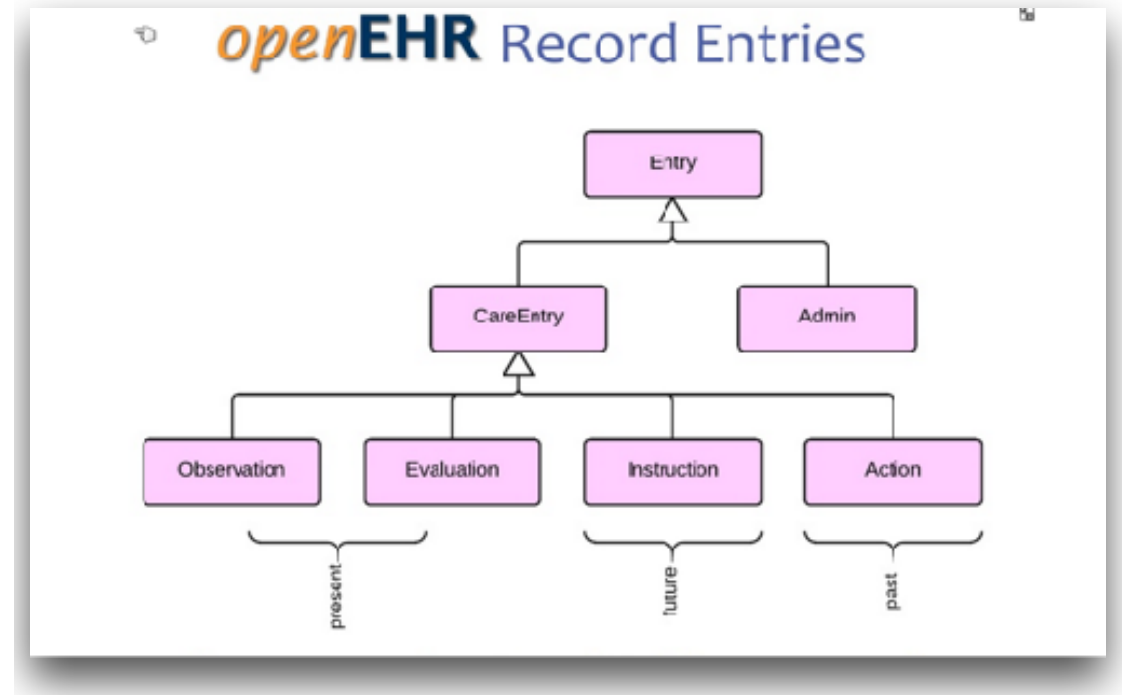
- Root 'document' for clinical data
- Carries most key medico-legal metadata composer (clinical_author), start_time, end_time organisation, clinical setting
- All recorded patient data saved inside a Composition Carries unique ID
- UID::serverID::Version_Suffix
- **5c8a8636-bc98-4441-abd5-e9cf396e8833::ripple_osi.ehrscape.c4h::1** Versioned
- All changes will create a new version

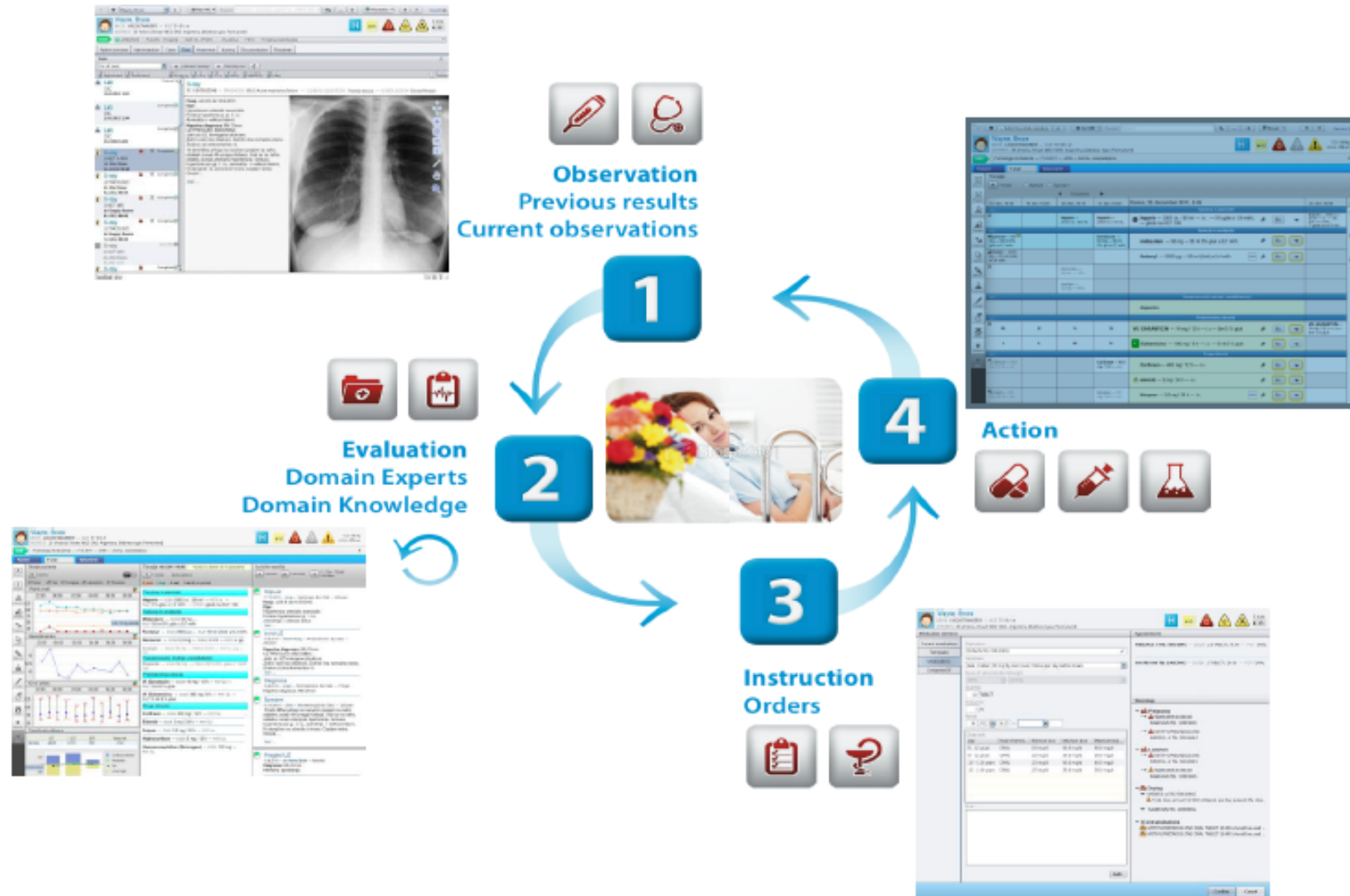
- **Composer**
- The clinical author identifier
- **Participations**
- Other individuals involved
- **Attestations**
- Sign-off by a senior clinician
- **Committer, commit_time**
- Person and Datetime that the record was 'saved'
- **Start_time, End_time**
- Start and end times of the clinical encounter
- **Health_care_facility, Location**
- Polyclinic identifier and specific location e.g. "Clinic Room 3"

- Used to divide complex compositions into manageable pieces
- Just for human navigation and organisation
- Can be nested
- No important clinical RM attributes



- A set of ENTRY sub-classes **carry all of the clinical payload**
- These are organised to fit the '**Clinical investigator**' cycle
 - OBSERVATION
 - EVALUATION
 - INSTRUCTION
 - ACTION
 - ADMIN ENTRY





- Use this video to discover the OpenEHR CKM:
- <https://www.youtube.com/watch?v=msW4Ctc7rv8>
- <https://ckm.openehr.org/ckm/>

Workflow

- <https://www.youtube.com/watch?v=B-W5c3qwpFE&t=207s>

- OpenEHR and FHIR
- READ

<https://www.youtube.com/watch?v=biEXVRzjWmw&t=355s>

- Archetype Query Language (AQL) is the query language of openEHR
- Unlike query languages, such as SQL or XQuery, AQL is a semantic query language rather like SPARQL expressing queries at the archetype level with a syntax that is independent of applications, programming languages, system environment, and storage models
- AQL queries can be run over multiple openEHR Clinical Data Repositories irrespective of their technology and internal physical data structures
- need to know what archetypes and templates have been registered with that system

- **Example:** select all nursing reports for a single patient with identifier 'a4a3f0c8-254a-49a5-97dd-784d50c46b78' during a period of an admission, and return the narrative text of the report 'synopsis', along with the report date.

```
SELECT s as Synopsis, c/context/start_time as reportDate
FROM EHR e[ehr_id/value='a4a3f0c8-254a-49a5-97dd-784d50c46b78']
  CONTAINS Composition c[openEHR-EHR-COMPOSITION.report.v1]
  CONTAINS
    Evaluation s[openEHR-EHR-EVALUATION.clinical_synopsis.v1]
WHERE c/name/value matches {'Nursing report'}
AND c/context/start_time >= '2014-04-19T00:00:00:00'
AND c/context/start_time < '2014-04-24T00:00:00:00'
```



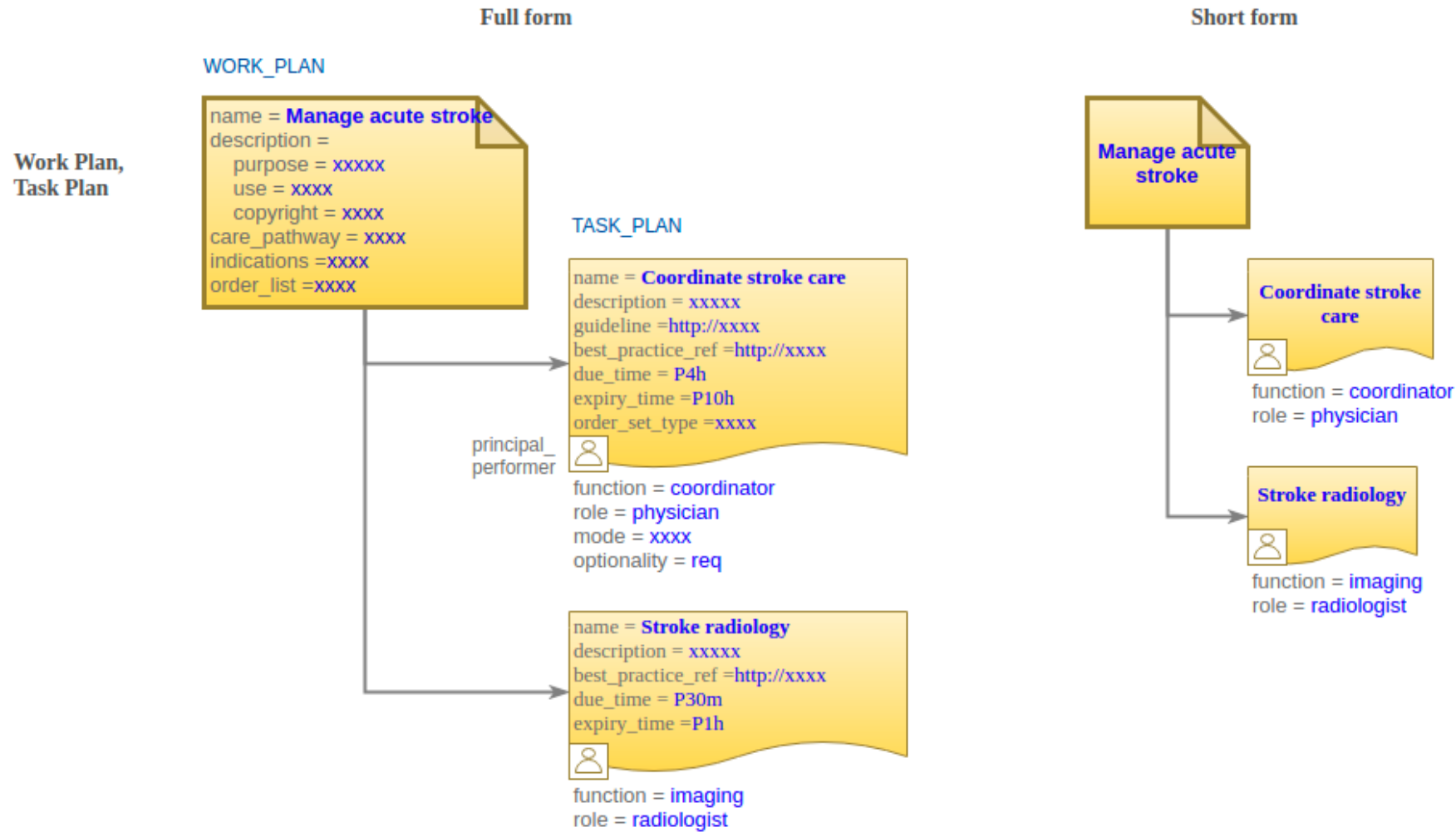
```
"resultSet": [  
  {  
    "reportDate": "2014-04-20T00:11:02.518+02:00",  
    "synopsis": "Pretty poorly overnight but better this morning"  
  },  
  {  
    "reportDate": "2014-04-21T00:11:02.518+02:00",  
    "synopsis": "Much better today. Possible discharge later"  
  },  
]
```

- openEHR provides the Guidelines Definition Language (GDL) as an extension to the openEHR standard. GDL allows the representation of clinical guidelines in an open and computable format.

<https://specifications.openehr.org/releases/CDS/latest/GDL.html>

- openEHR provides the openEHR Task Planning Visual Modelling Language (TP-VML) to enable workflow and task planning to be managed. This is the most recent extension to the openEHR specification.

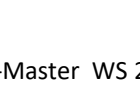
https://specifications.openehr.org/releases/PROC/latest/tp_vml.html



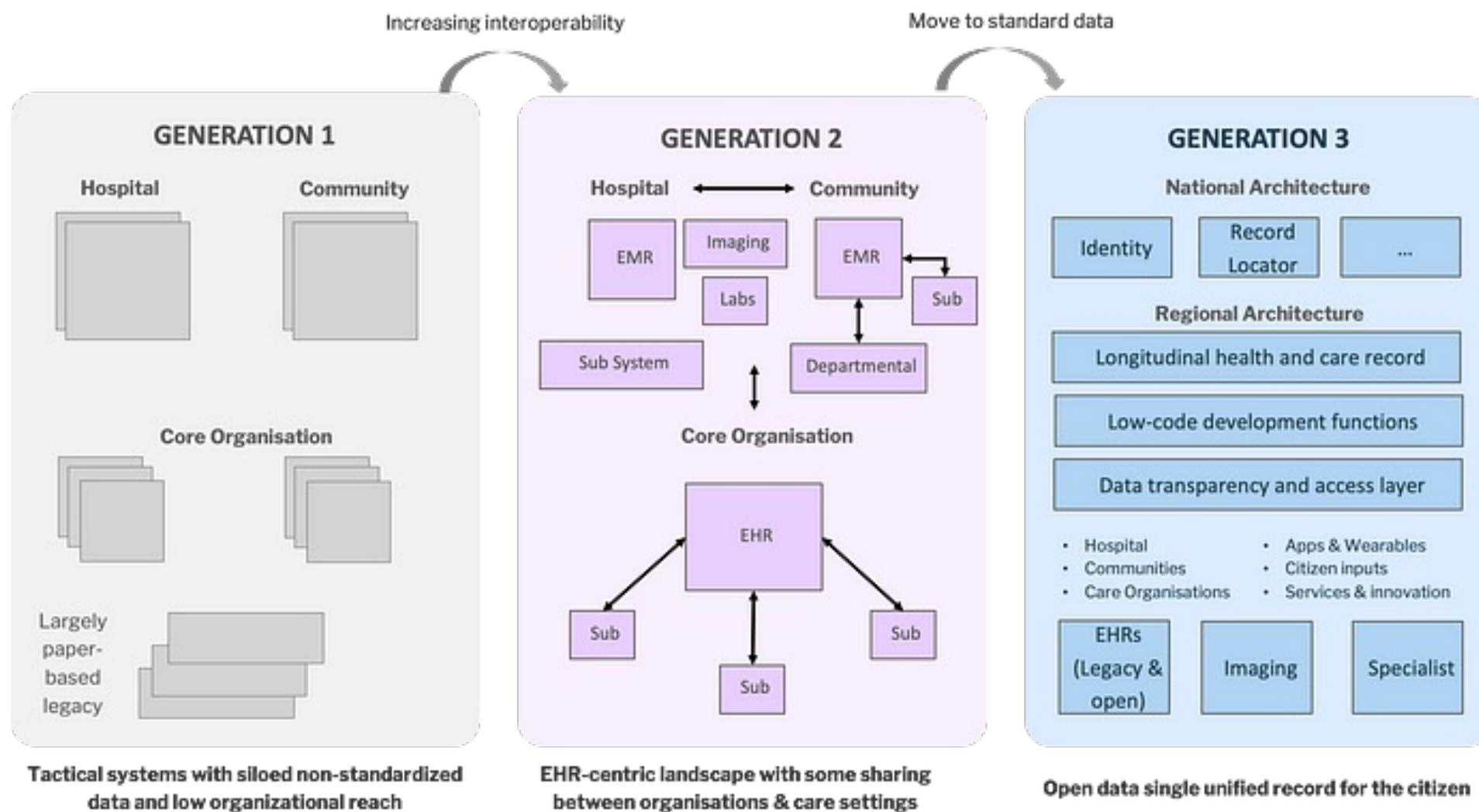
Modelling Tools

<https://www.openehr.org/downloads/modellingtools/>

Free to use & open source

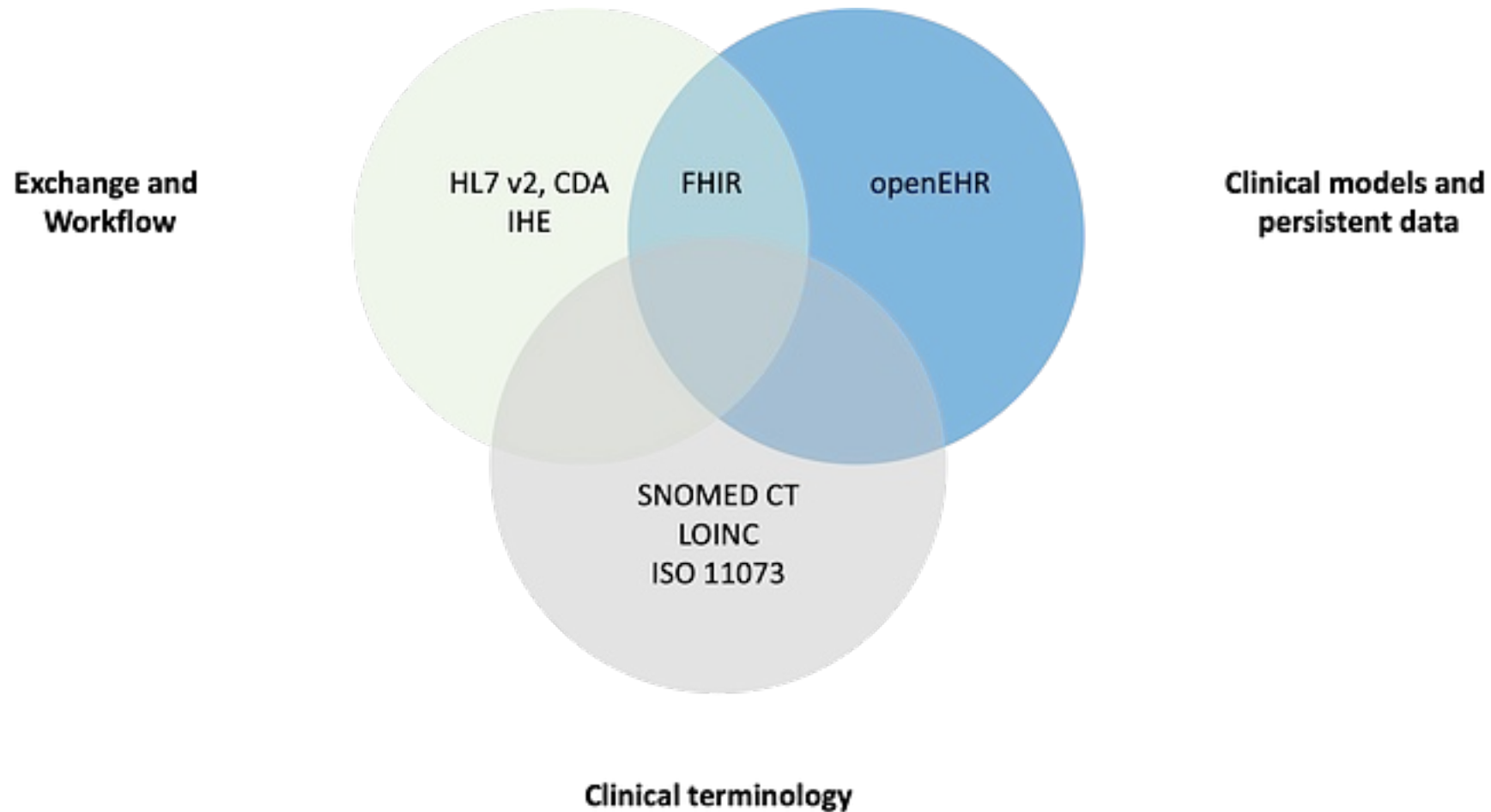
Product	Description	Artefacts	Platform	Licence	Resources
 ADL Workbench	Reference model-driven visual IDE for parsing, compiling, analysing, converting and editing archetypes and templates. Built on the reference ADL parser.	ADL14 archetypes, ADL2 archetypes, ADL2 OPTs, BMMs.	Windows, Linux	Apache2	Project website Support User manual
 Archetype Designer	openEHR Archetype and Template visual design tool.		All		Archetype Designer
 Archetype Editor	The Archetype Editor is currently the main tool in use for authoring archetypes as found on openEHR CKM and elsewhere. It is Unicode-enabled and works with archetypes in any language. The editor application has been localised to several languages, including Danish, English, Farsi, German, Japanese, Russian, Spanish, Swedish and Turkish.	ADL 1.4 archetypes	Windows	Mozilla MPL	Project website Support User manual
 LinkEHR	The LinkEHR Editor works with multiple reference models and languages. Reference models can be plugged in in XML-schema or openEHR BMM (basic meta-model) format.	.oet templates; ADL 1.4 OPTs; ADL 1.4 archetypes.	All		LinkEHR Support
 Ocean Health Systems	The Template Designer is the tool required for editing '.oet' (pre-ADL2) templates.	.oet templates	Windows		Template Designer

OPENEHR AND FHIR



Source: Professor Rachel Dunscombe NHS Digital Academy

- **Generation 1** — A largely paper-based landscape, with some tactical systems used within certain organisations. Data is typically held in a non-standardised way and there is limited sharing of information.
- **Generation 2** — Generation 2 systems are centred around the EHR. These systems are often closed but there is some sharing of information between organisations and care settings. Information is typically shared by moving information around using messages and occasionally APIs.
- **Generation 3** — This sees a shift to a regional model where systems and processes are designed around the citizen — not the organisation or EHR. At the core of this is a move to an open, standardised data record for the citizen. One that can be used in a safe, governed way by applications across all organisations and care settings — irrespective of where the person is.



Source: Professor Rachel Dunscombe NHS Digital Academy

Use Case Summary

openEHR optimized for:

- Storage of data in vendor neutral format
- Clinically designed models following "maximal dataset"
- Semantic querying of data through AQL

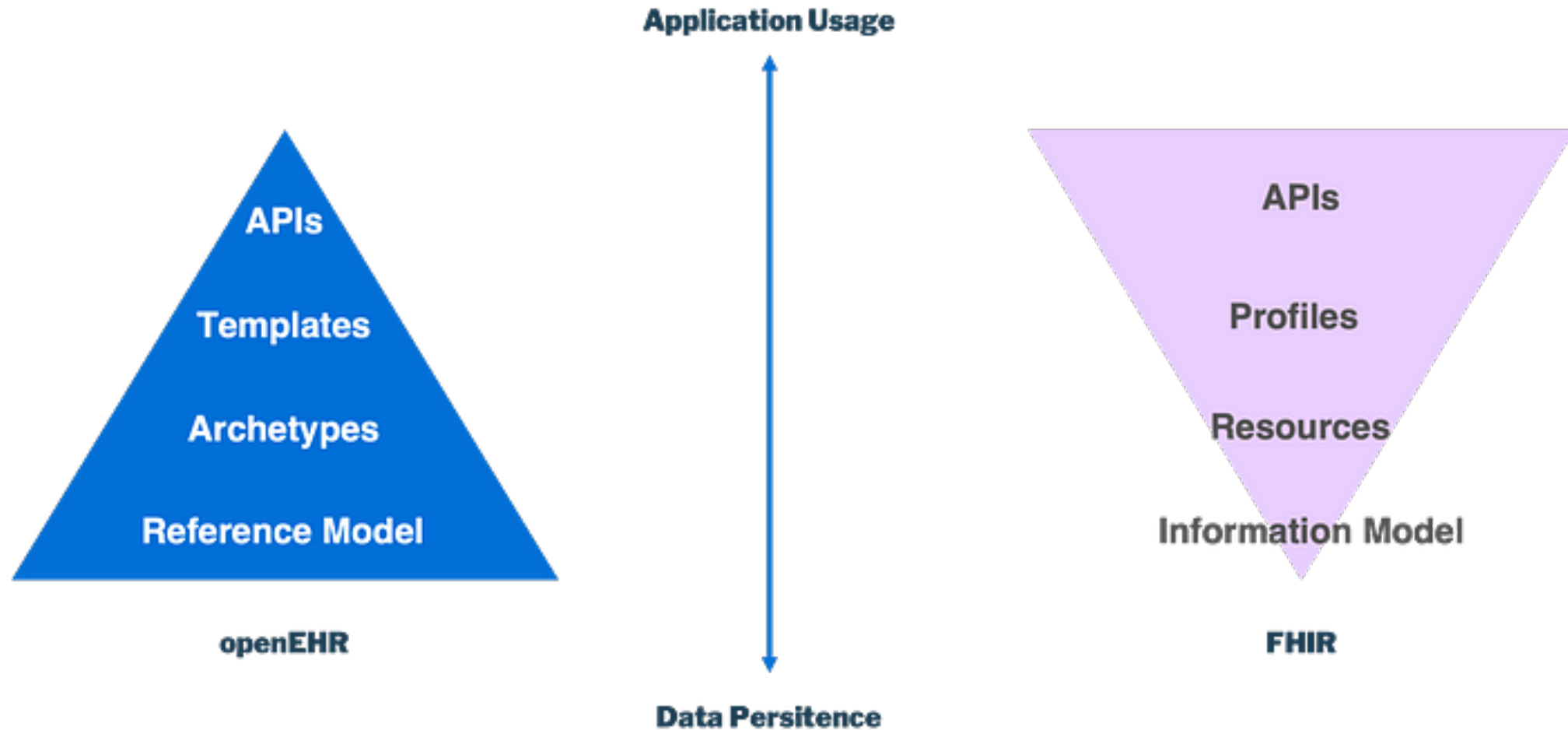
*open***EHR**

FHIR optimized for:

- Exchange of information between systems
- Common models following "80/20" rule
- Exposing data through industry adopted API's



- FHIR has an 80/20 rule designed to be extended or customized for specific use cases
- openEHR has a maximal data model designed to be constrained for specific use cases
- Both standards might be an invert from each other



“The first law of software architecture is everything is a trade-off”.

- June 2022:
- FHIR (v4.0.1) has 145 resources
- of which 11 are classified at the Normative Level of stability and implementation readiness
- Normative resources are considered to be stable and locked
- not normative are typically either classified as Draft or Trial Use
- Trial Use resources have been reviewed but have not yet seen widespread production use --> *“Future versions of FHIR may make significant changes to Trial Use content that are not compatible with previously published content.”*
- 92% of resources in the current published version of FHIR are Trial Use and come with a compatibility health warning

- Each time you upgrade to a new version of FHIR you will need to review the changes and assess the impact on your deployment
- From name change to information model change
- E.g. AllergyIntolerance “assertedDate” changed to “recordedDate”
 - Name and semantics changed
- openEHR has been designed to change more easily
- base reference model
- versioning of openEHR artefacts (archetype, template, terminology subset) can be applied to each individual source artefact, rather than the entire repository

- FHIR extensions and profiles are a necessary, but complicated part of FHIR
- FHIR resources need to be extended or profiled when dealing with use-case specific data that is not covered by the core published resource
- Unless you have a really basic use case a reasonable assumption is that you will require extensions
- great flexibility for implementers, but it also creates a number of challenges
- significant variation in terms depending if native FHIR and local or national profiles

“HL7 FHIR is not a standard, it is a standards building framework.”

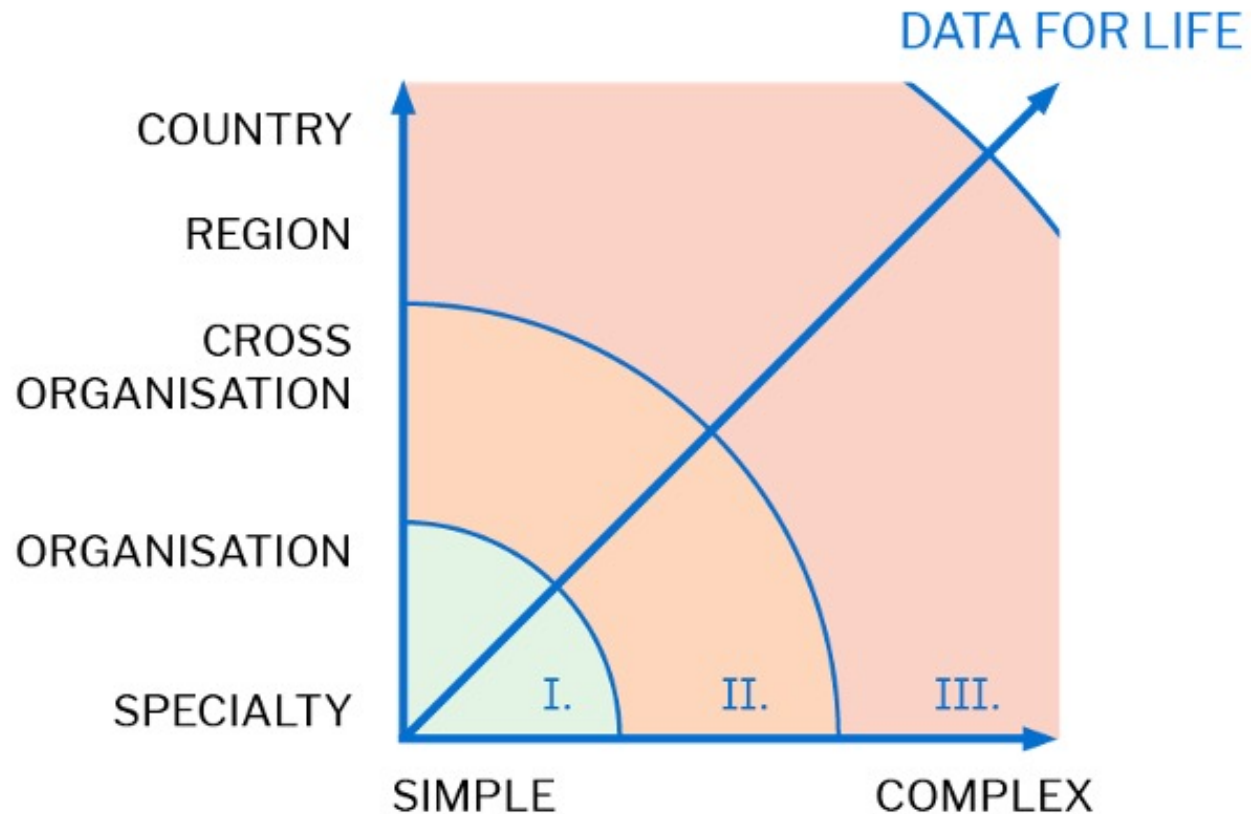
- the profiles or implementation guides are the standard in fact
- local version of a standard → define conflicting ways to store the data
- openEHR addresses this challenge from the other direction
- Instead of having a narrow set of models (i.e. resources) that get extended for each use case (via profiles and implementation guides), it adopts the idea of a “maximal data model” with a large core of mature models (archetypes) that are widely understood
- subsequently get constrained at runtime (via templates)

- “80/20” rule → FHIR is focused on 80% of the common use cases
- several resources are modelled at a higher level of abstraction from the subject area being modelled and stored
- complexity “leaks” through the abstraction
- For example ONE observation resource
- openEHR each of the different types of observation is modelled around the full spectrum of observation types such as Blood Pressure, Heart Rate, Body Temperature etc
- questionnaire resources → “anti-patterns” emerge
- creates barriers when it comes to understanding what this information actually represents

- Needs to be considered when sharing outside of organizational boundaries
- And when stored for long term usage

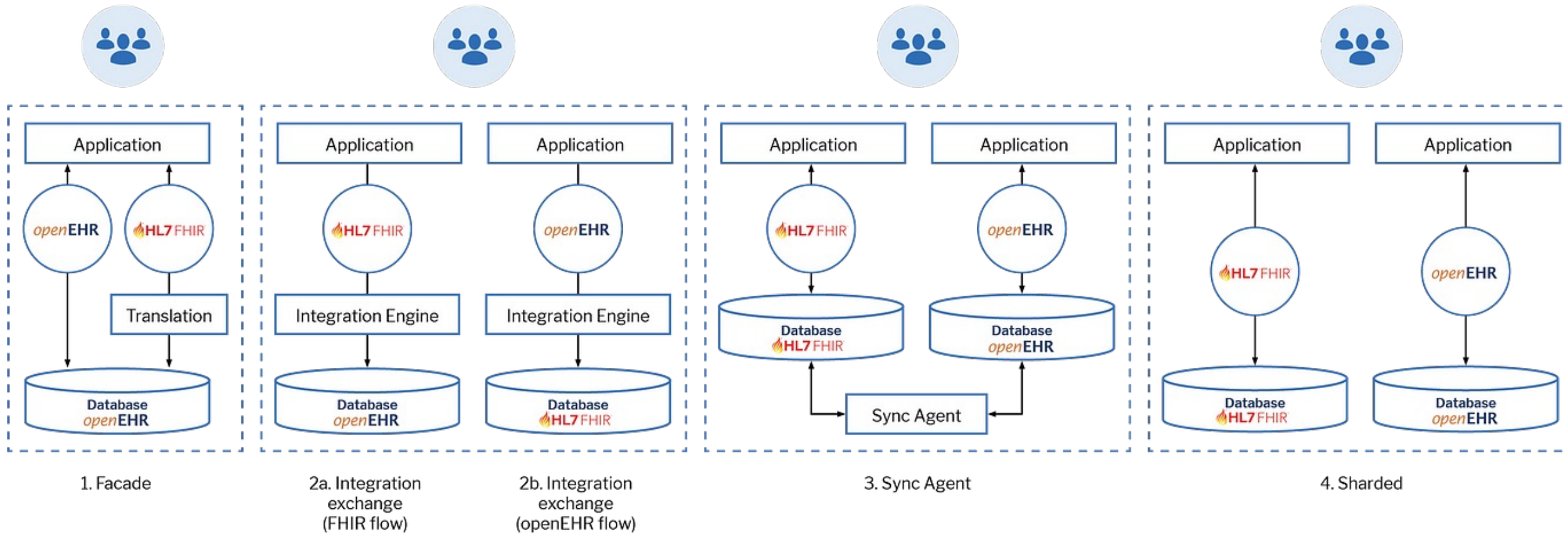
- Patient resource -> 23 search parameters are (currently) classified as Trial Use
- FHIR Search can quickly become complex → “joins” across resources, composite search (multiple query parameters) or chained search (traversing references in the context of a query)
- In openEHR native query language called AQL

- trade-offs outlined above can reduce overall development productivity and agility
- smaller number of models more time needs to be spent modelling and profiling Resources
- openEHR: Clinical Knowledge Manager (CKM) to collaborate, 100 countries — where over 1500 clinicians have helped curate over 800 clinical models



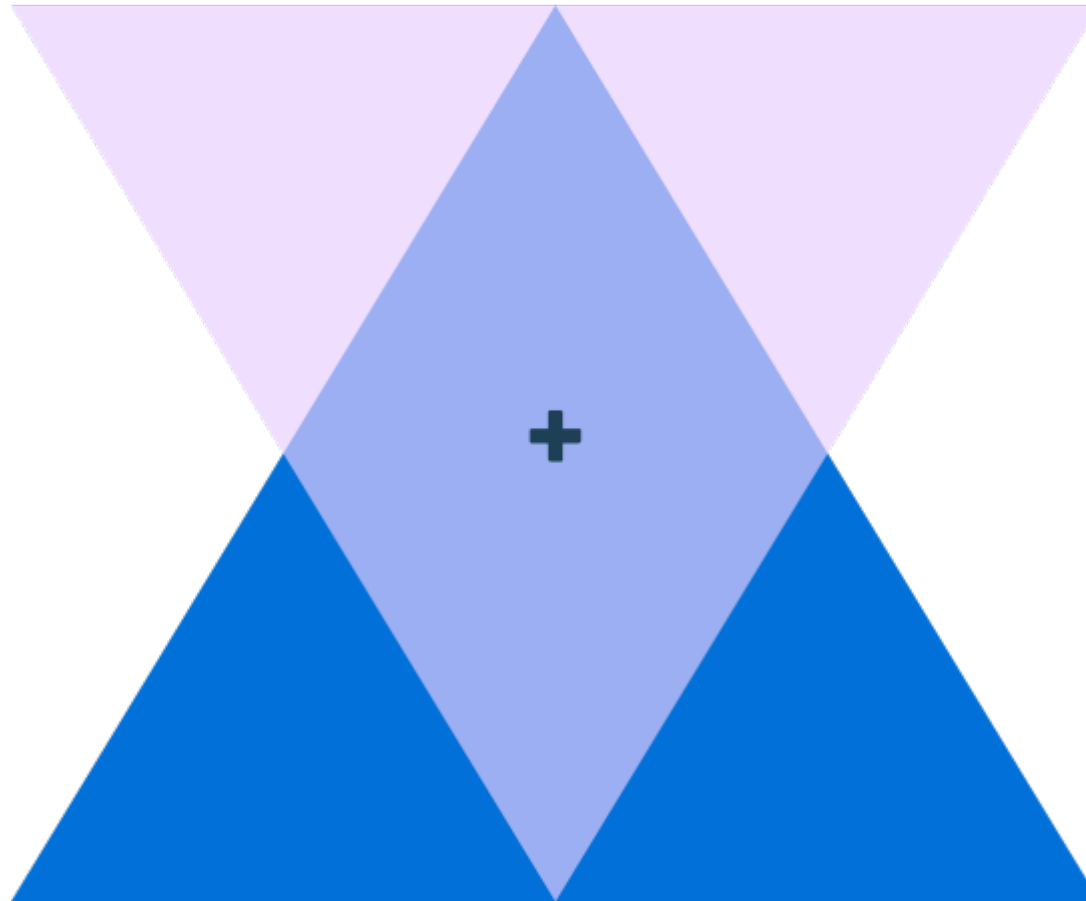
- I. Safe zone — FHIR as a persistence mechanism will likely work well
- II. Danger zone — Requires IT oversight, governance and appropriate strategies to mitigate trade-offs
- III. Off limits — Adopt FHIR for exchange but not persistence

- combination of standards is required in order to help solve the complex problems in healthcare
- care to be delivered outside of the hospital and closer to the patient
- emergence of vast new streams of data



- **Pattern 1- Facade** — A translation component is introduced to translate inbound and outbound FHIR requests to an openEHR repository. Typically alongside this, openEHR APIs will be used to support a complete set of functionality.
- **Pattern 2 — Message Broker** — An integration engine acts as a message broker to translate incoming messages from external applications. 2b is shown for completeness but it is not a practical use case

- **Pattern 3 — Sync Agent** — A sync agent exists to copy a defined group of data items between a FHIR and openEHR repository.
- **Pattern 4 — Sharded** — FHIR and openEHR exist alongside each other with data being stored in each location based on the nature of the data. For example to use FHIR to store much of the administrative data, including patient demographics, encounters and appointments.
- Translation challenging still today - mappings within a proprietary integration engine



*open*EHR